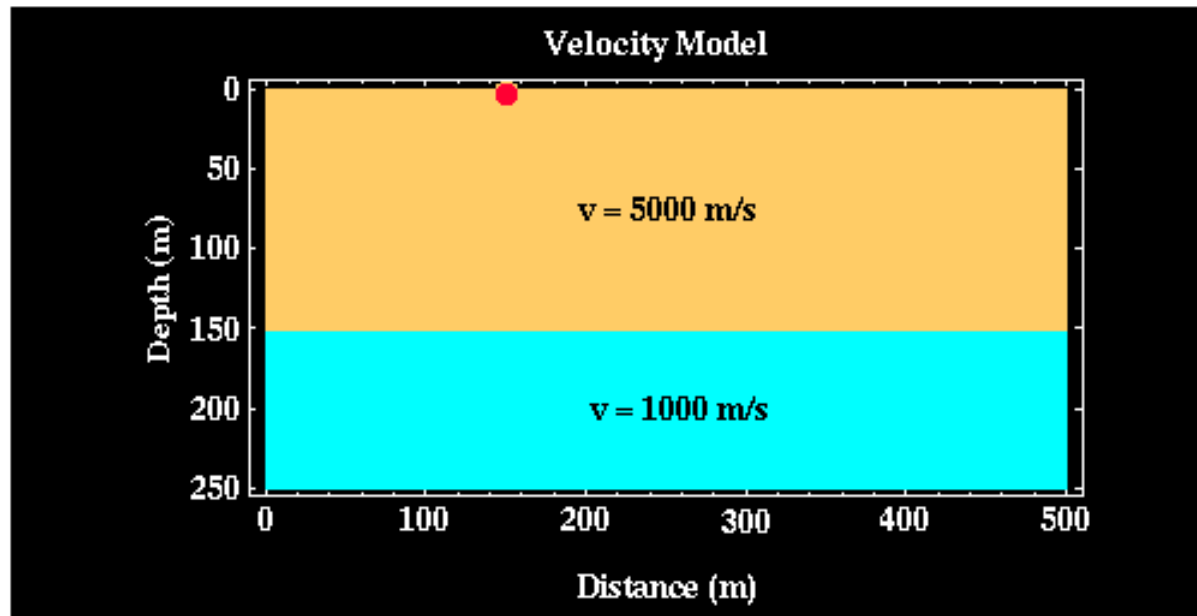
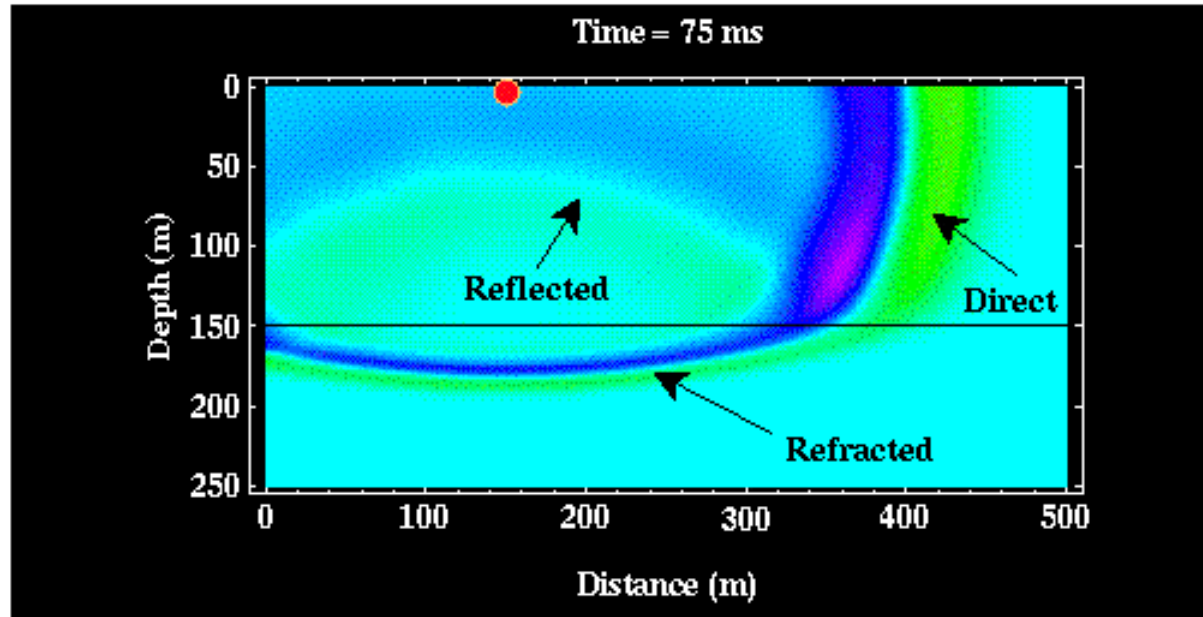


Travel Times: High Velocity Layer Over Low Velocity Halfspace

Now, the [first model](#) we considered consisted of a high velocity layer over a lower velocity half-space, as shown below.



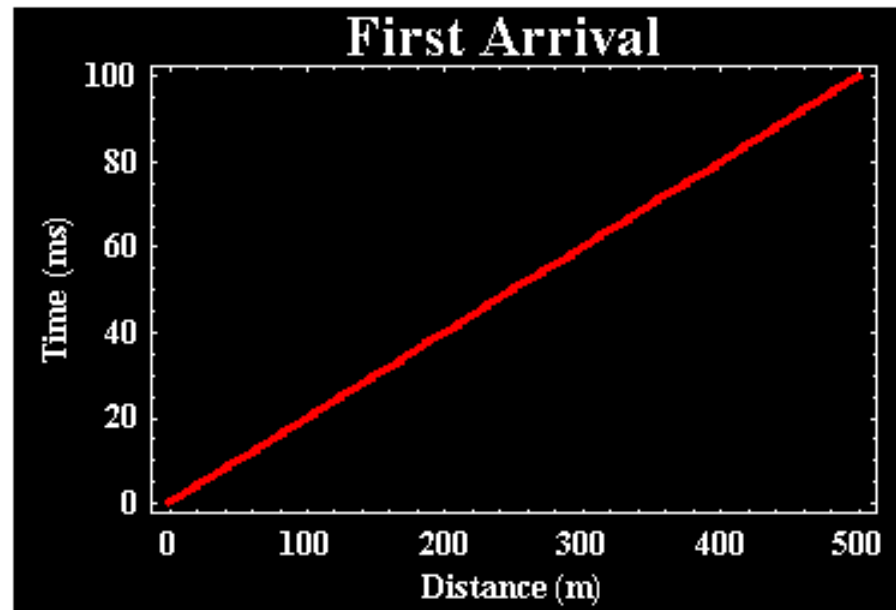
For this model, what first arrivals would you expect to see, and what can you determine about the subsurface structure from these arrivals? A snapshot of the waves produced from a surface source as they interact with the boundary is shown below. This is the same image that was shown [previously](#).



As was described earlier, we need to consider three wave types. The direct arrival, the reflected arrival, and the transmitted (or refracted) arrival. Notice that the difference between the waves produced in this model and those produced when a low velocity layer overlies a higher velocity halfspace is the absence of a head wave.

Let's consider the transmitted (refracted) arrival first. The refracted arrival propagates downward from the boundary. If there is no other structure below our first layer, this wave will continue to move downward. As such, it can never be observed by receivers located on the Earth's surface.

The reflected arrival can be observed on the Earth's surface, but it must always arrive after the direct arrival. Thus, as before, the reflected arrival is *never* a first arrival, and we therefore do not use it in refraction surveying. The only arrival we'll observe as a first arrival is, therefore, the direct arrival. In this model, the direct arrival propagates away from the source at constant speed (5000 m/s). So, if we were to plot the time of arrival of the first arrivals versus distance from the source, we would get a figure like the one shown below.



What can we learn from this plot? Well, not too much. From the plot we could compute the speed at which seismic energy travels in the layer, just as we did [previously](#). But notice that our first arrival information no longer tells us anything about the speed at which seismic waves travel in the halfspace. In fact, they no longer indicate any existence of the halfspace!! By this, I mean the travel times shown above would be identical for a uniform Earth that had a wave propagation speed of 5000 m/s.

This example illustrates one of the fundamental limitations of using the seismic refraction method. To be successfully employed (i.e., to get the correct answers from the recorded data), seismic wave speed *MUST* increase with depth. If wave speed decreases with depth, you will not be able to detect this wave speed decrease. And you will, as a consequence, almost assuredly interpret the first arrivals incorrectly which would result in an estimate of the subsurface structure that is wrong.

Seismic time-distance curves from the geological model shown for both reflection and refraction events.

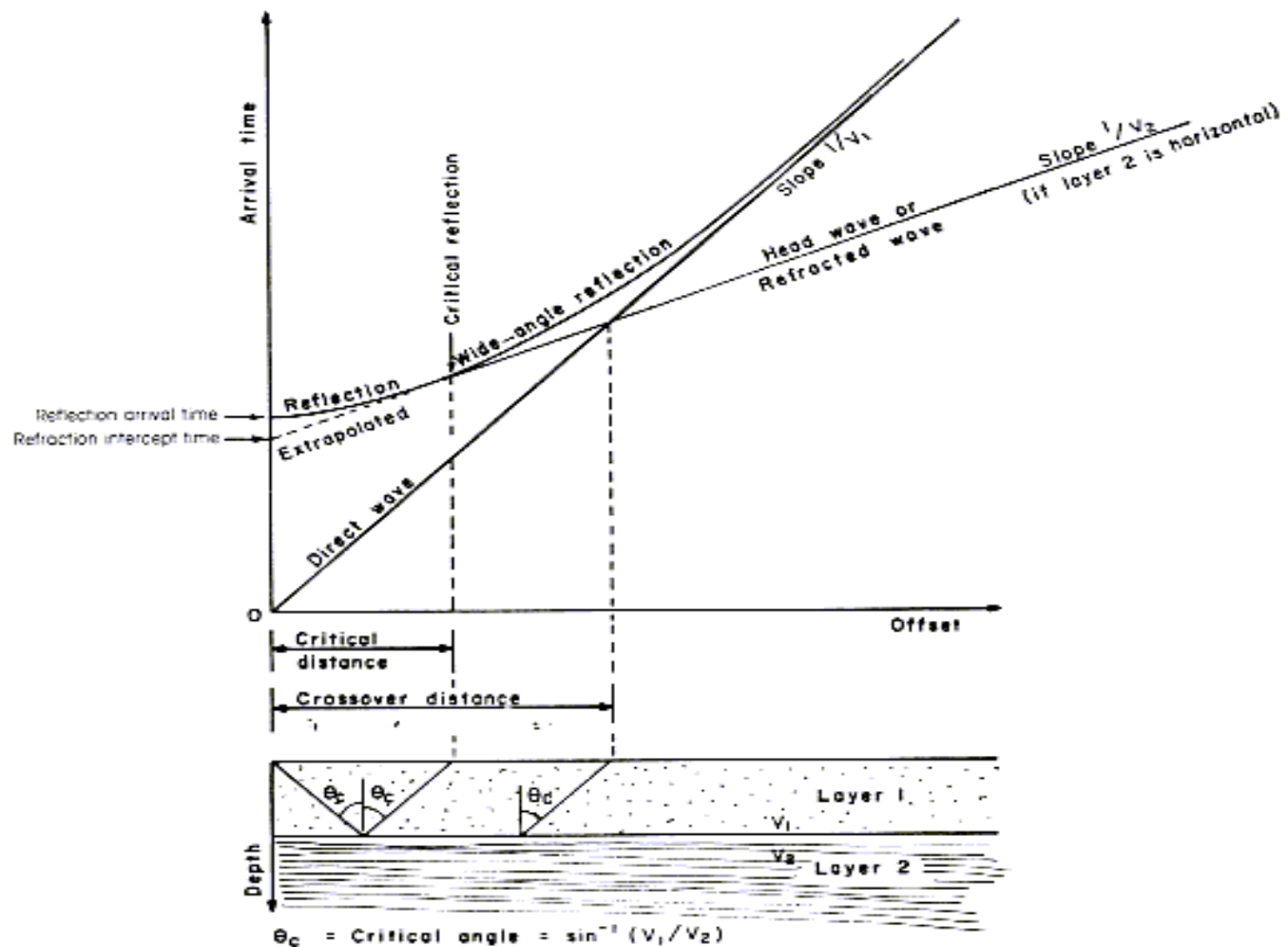
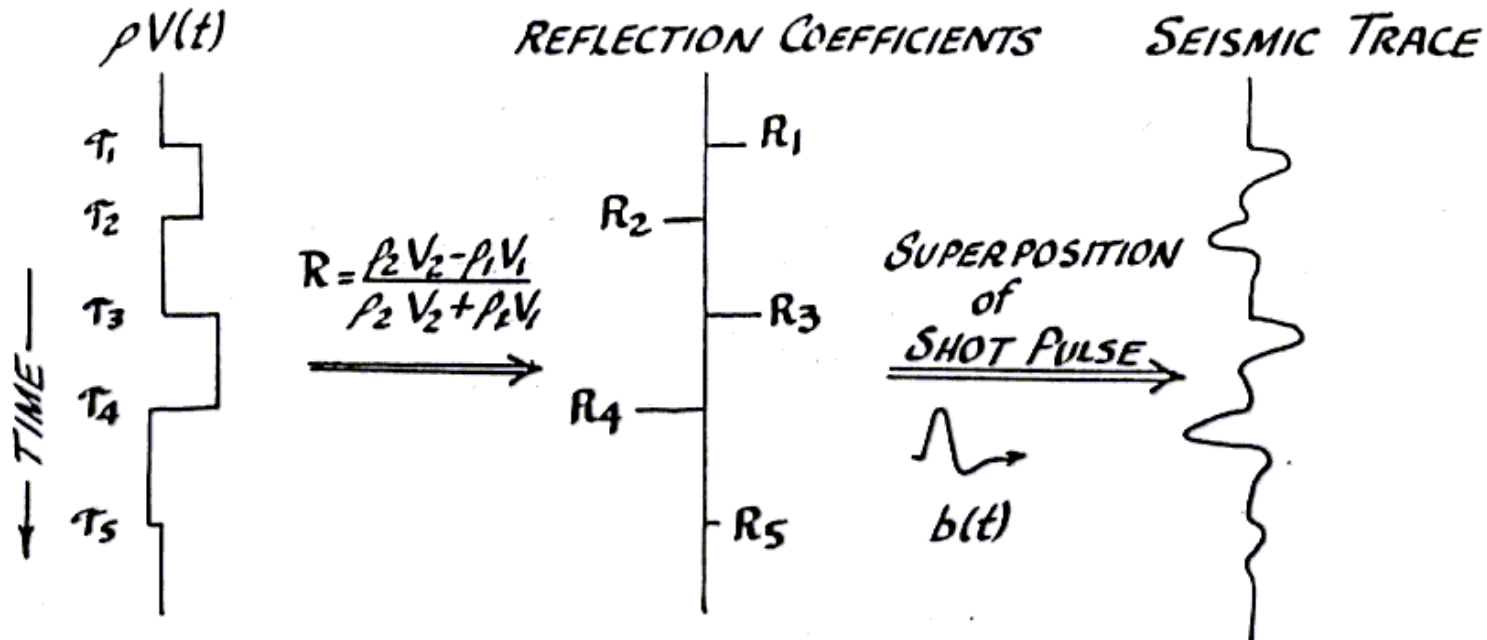


FIG C-15. **Critical distance.** Curvature of the reflection is hyperbolic if velocity above reflector is constant. Reflection amplitude may be exceptionally large in the vicinity of critical reflection.

Schematic overview of the convolutional model of the seismic reflection method.



$$S(t) = R_1 b(t-\tau_1) + R_2 b(t-\tau_2) + R_3 b(t-\tau_3) + R_4 b(t-\tau_4) + R_5 b(t-\tau_5)$$

SEISMIC TRACE = WEIGHTED and DELAYED SUPERPOSITION of SHOT PULSE.

$$S(t) = R(t) * b(t)$$

From optics, the concept and use of the source image.

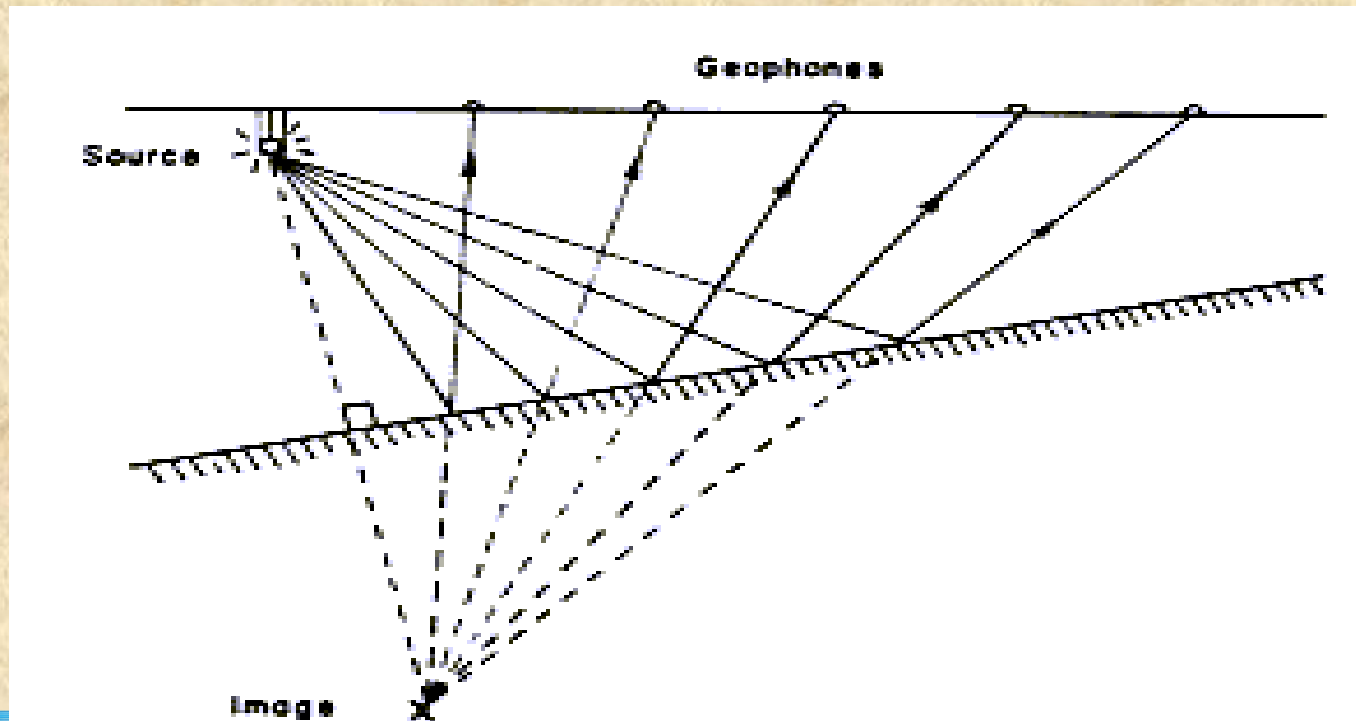
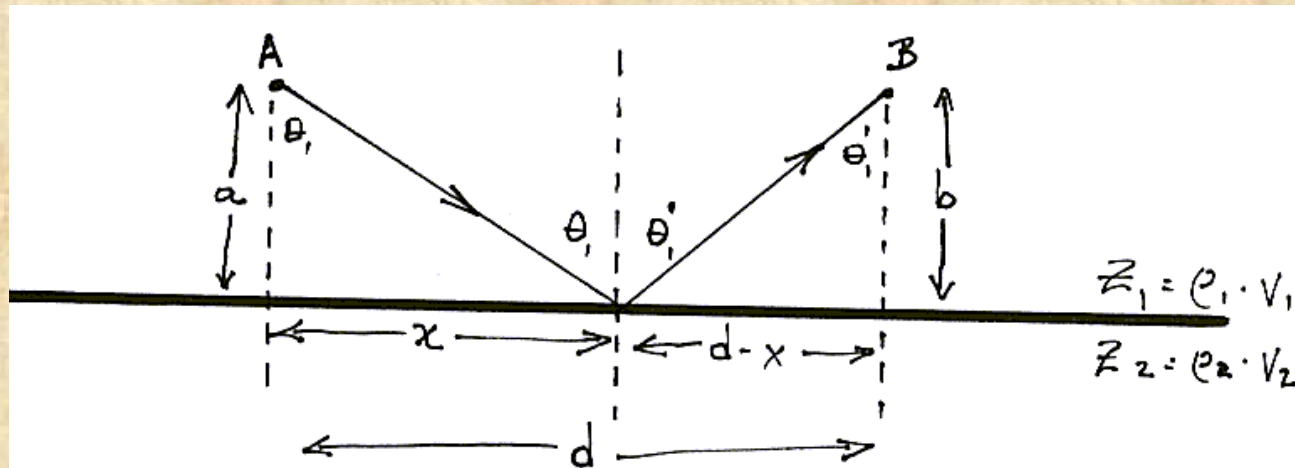


FIG. I-1. Reflected rays reaching the geophones from the source appear the same as direct rays from the **image point**. The use of straight raypaths implies constant velocity.

Fermat's Least-Time Principle

A light ray traveling from one point to another will follow a path such that, compared with nearby paths, the time required is either a minimum or a maximum or will remain unchanged (it [the ray] will remain stationary).

Using Fermat's Least-Time Principle to prove the relationship of incident to reflected waves.



total length of Ray $l = \sqrt{a^2 + x^2} + \sqrt{b^2 + (d-x)^2}$

to minimize the length we set $dl/dx = 0$

thus $dl/dx = \frac{1}{2}(a^2 + x^2)^{-1/2}(2x) + \frac{1}{2}[b^2 + (d-x)^2]^{-1/2}(2)(d-x)(-1) = 0$

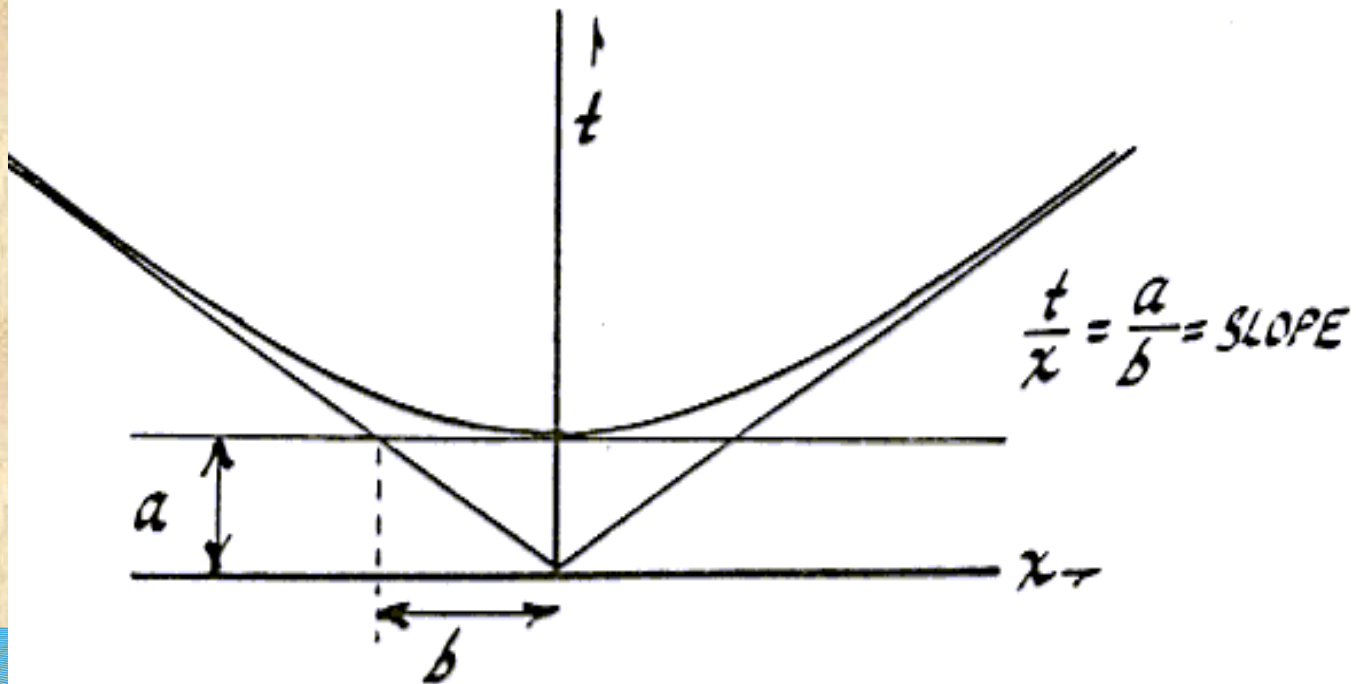
OR $\frac{x}{\sqrt{a^2 + x^2}} = \frac{d-x}{\sqrt{b^2 + (d-x)^2}}$

which is $\sin \theta_i = \sin \theta_r \Rightarrow \theta_i = \theta_r$



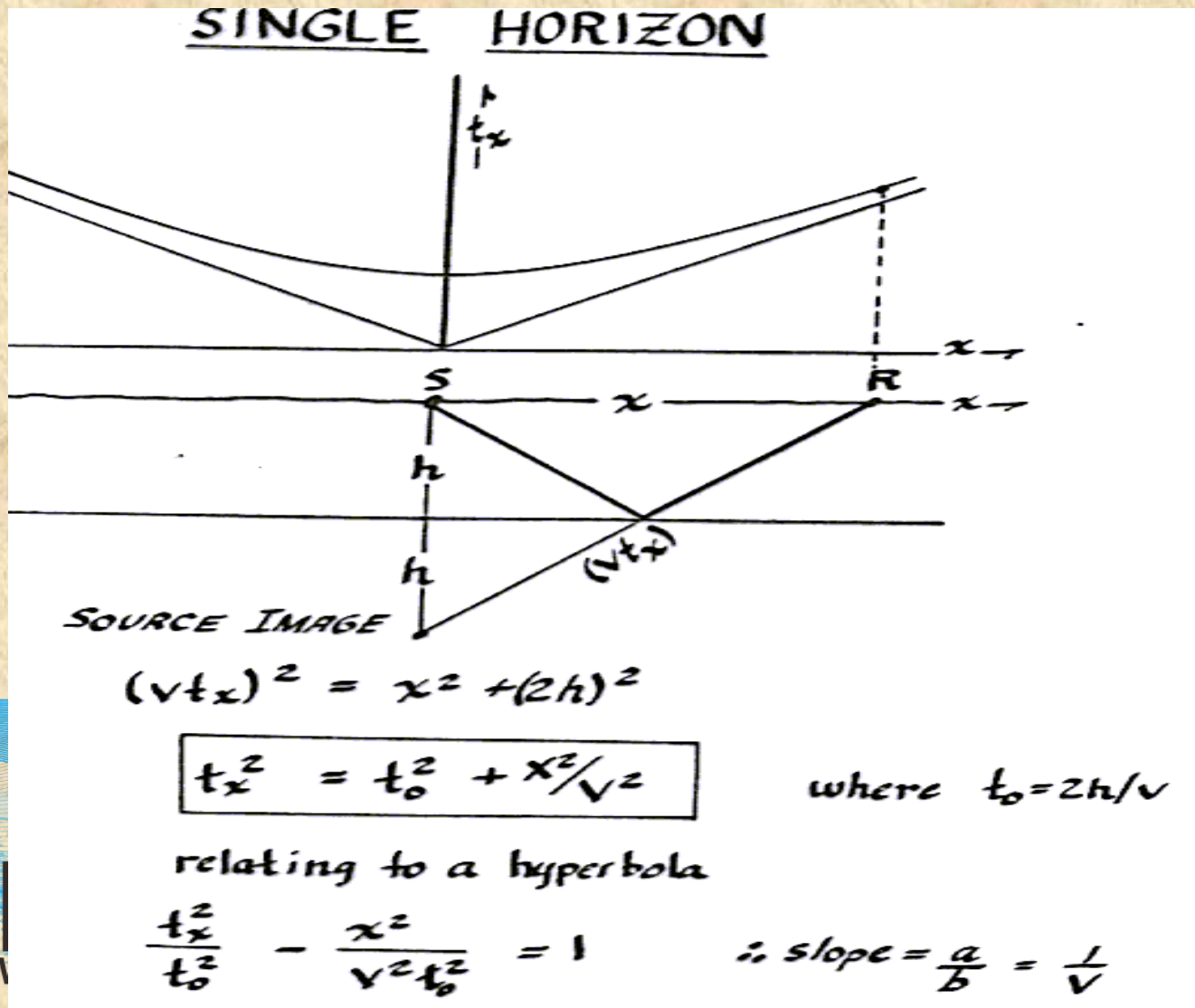
Since we are going to discuss reflection hyperbolae, then a review of the analytical equation for a hyperbola is in order.

REVIEW of HYPERBOLA



$$\frac{t^2}{a^2} - \frac{x^2}{b^2} = 1$$

Here the analytical time-distance solution for a single reflection event is related back to the general expression of the hyperbola.



Derivation of reflection equation for single boundary

Derive Reflection Equation Single Horizon

Pythagoras Theorem gives:

distances $\rightarrow$ (1) $(\bar{v} t_x)^2 = x^2 + (2h)^2$

Divide (1) by $(\bar{v})^2$ gives

times $\rightarrow$ (2) $(t_x)^2 = \frac{x^2}{\bar{v}^2} + \frac{(2h)^2}{\bar{v}^2}$

Now $\left(\frac{2h}{\bar{v}}\right)^2$ is a special case of $t_x @ x=0$
i.e. source & receiver at same spot $\Rightarrow$ re-phrased $(t_0)^2$

thus (2) becomes $(t_x)^2 = \frac{x^2}{\bar{v}^2} + (t_0)^2 \Leftarrow$

If we have a discrete layer
where the velocity is constant then
 $\bar{v}$ becomes the true velocity for layer

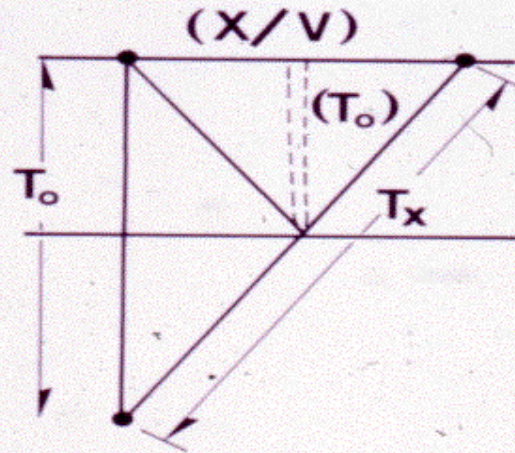
where $\bar{v} \equiv$ average velocity
that ray path has over
the distance $\bar{v} t_x$

AND $t_x \equiv$ recorded travel
time from shot
to detector at x

Here is the equation that
all geophysicists know
AND, as we will see, will hold
for multiple layers replacing
 $\bar{v}$ with V_{rms} (root-mean-
squared) velocity



From the theorem of Pathagoris, comes the expression for normal move out (NMO) for non-normal incident reflection.



Analytic Expression of NMO - No Dip

NMO = Time Correction Due to Shot - Receiver Separation

$$\Delta t = T_x - T_o$$

$$T_x^2 = T_o^2 + (X/V)^2$$

Exact Expression of NMO

$$\Delta t = \sqrt{T_o^2 + (X/V)^2} - T_o$$



Normal Moveout Equation

Normal moveout is applied according to the following formula:

$$T_x = \sqrt{T_0^2 + \left(\frac{X}{V}\right)^2}$$

T_x = The actual reflection time of the seismic event due to NMO effects

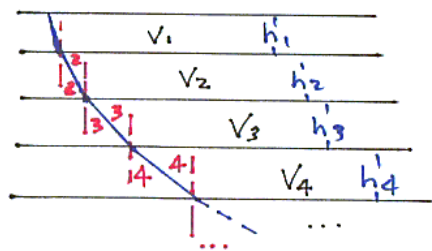
T_0 = The zero offset reflection time of the seismic event

X = The actual source-receiver offset distance

V = The NMO or stacking velocity for this reflection event.



Derivation of reflection equation for multiple boundaries



Reflection Equation
multi-layer (Parametric)

$$P \equiv \text{RAY parameter} = \frac{\sin \theta_1}{v_1} = \frac{\sin \theta_2}{v_2} = \frac{\sin \theta_3}{v_3} = \dots$$

P is generalized

Swell's law operator

Now if $p = \frac{\sin \theta_i}{v_i}$

then $\sin \theta_i = p v_i$

AND $\cos \theta_i = (1 - p^2 v_i^2)^{1/2}$

AND $\tan \theta_i = \frac{p v_i}{(1 - p^2 v_i^2)^{1/2}}$

times

→ So $t_x = 2t_1 + 2t_2 + 2t_3 + \dots + 2t_N$

then

$$t_x = \frac{2h_1}{v_1 \cos \theta_1} + \frac{2h_2}{v_2 \cos \theta_2} + \dots + \frac{2h_N}{v_N \cos \theta_N}$$

2 because
2-way travel time

OR $t_x = 2 \sum_{i=1}^N \frac{h_i}{v_i (1 - p^2 v_i^2)^{1/2}} \quad (\text{using } p)$

Likewise we have distances

$$x = 2\Delta x_1 + 2\Delta x_2 + \dots + 2\Delta x_N$$

OR $x = 2h_1 \tan \theta_1 + 2h_2 \tan \theta_2 + \dots + 2h_N \tan \theta_N$

* OR $x = 2 \sum_{i=1}^N \frac{h_i p v_i}{(1 - p^2 v_i^2)^{1/2}} \quad (\text{using } p)$

Derivation of reflection equation for multiple boundaries (cont.)

$t(x)$ time that is a function of distance

Reflection Equation
multi-layer (Taylor Series Expansion)

$t_x \approx t_0 + \frac{dt_x}{dx} \bigg|_{x=0} x + \frac{d^2 t_x}{2! dx^2} \bigg|_{x=0} x^2 + \dots$

0 (due to symmetry of hyperbola)

leaves $t_x \approx t_0 + \frac{d^2 t_x}{2! dx^2} \bigg|_{x=0} x^2$

now other terms beyond accuracy

Squaring both sides

$(t_x)^2 \approx (t_0)^2 + 2t_0 \frac{d^2 t_x}{2! dx^2} \bigg|_{x=0} x^2 + \left[\frac{d^2 t_x}{2! dx^2} \bigg|_{x=0} x^2 \right]^2$

dimensional analysis shows to be insignificant

Now $\frac{d^2 t_x}{dx^2}$ can be replaced by dp/dx since $p = \frac{\sin \theta_i}{v_i} = \frac{dt_x}{dx}$

So $(t_x)^2 \approx (t_0)^2 + \frac{2t_0}{2!} \frac{dp}{dx} \bigg|_{x=0} x^2$

Now the reciprocal of dp/dx namely $\frac{dx}{dp}$ can be determined from our parametric expression for $x = f(p)$

Derivation of reflection equation for multiple boundaries (cont.)

Reflection Equation

(cont.)

Multi-layer (Taylor Series)

$$\begin{aligned} \text{thus } \frac{dx}{dp} &= 2 \sum_{i=1}^N \frac{h_i v_i}{(1-p^2 v_i^2)^{1/2}} + 2 \sum_{i=1}^N p h_i v_i (-1/2) (1-p^2 v_i^2)^{-3/2} (-2 v_i^2 p) \\ &= 2 \sum_{i=1}^N \frac{h_i v_i}{(1-p^2 v_i^2)^{1/2}} + 2p \sum_{i=1}^N \frac{h_i v_i (-1/2) (2 p v_i^2)}{(1-p^2 v_i^2)^{3/2}} \end{aligned}$$

Now evaluating $\frac{dx}{dp}$ @ $x=0 \Rightarrow$ note At $x=0$ the ray path is normal (vertical)
therefore $\theta_i = 0 \Rightarrow \sin \theta_i = 0 \Rightarrow p = \frac{\sin \theta_i}{v_i} = 0$

$$\text{therefore } \left. \frac{dx}{dp} \right|_{x=0} = 2 \sum_{i=1}^N h_i v_i = 2 \sum_{i=1}^N \Delta t_i v_i^2 \quad \text{Substituting } h_i = \Delta t_i \cdot v_i$$

Substituting $\left. \frac{dx}{dp} \right|_{x=0}$ into our Taylor Series we get

$$(t_x)^2 = (t_0)^2 + \frac{t_0 x^2}{2 \sum_{i=1}^N \Delta t_i v_i^2} \quad \text{OR} \quad (t_x)^2 = (t_0)^2 + \frac{x^2}{(V_{\text{RMS}})^2} \quad \text{where}$$

$$\frac{1}{(V_{\text{RMS}})^2} = \frac{t_0}{2 \sum_{i=1}^N \Delta t_i v_i^2} = \frac{t_0/2}{\sum_{i=1}^N \Delta t_i v_i^2} \quad \text{OR} \quad V_{\text{RMS}}^2 = \sum_{i=1}^N \Delta t_i v_i^2 / t_0/2$$

For the multi-layered reflection situation, the problem reduces to a Taylor's Series expansion.

MULTI-LAYED REFLECTION

(TAYLOR SERIES EXPANSION)

$$(1) \quad t_x = t_0 + \left. \frac{dt_x}{dx} \right|_{x=0} x + \frac{d^2 t_x}{2! dx^2} \bigg|_{x=0} x^2 + \dots$$

ODD DERIVATIVE TERMS = 0 (SYMMETRY)

$$(2) \quad t_x^2 = t_0^2 + \frac{2t_0}{2!} \frac{d^2 t_x}{dx^2} \bigg|_{x=0} x^2 + \dots$$

BECAUSE $p = \sin \theta_i / v_i = dt_x / dx$, $dp/dx = d^2 t_x / dx^2$
FROM PARAMETRIC FORM OF x WE OBTAIN

$$(3) \quad \frac{dx}{dp} = 2 \sum^n \frac{h_i v_i}{(1-p^2 v_i^2)^{1/2}} + 2p \sum^n \frac{h_i v_i (-1/2)(2p v_i^2)}{(1-p^2 v_i^2)^{3/2}}$$

EVALUATING (3) AT $x=0$ (ie $p=0$),

$$(4) \quad \frac{dx}{dp} = 2 \sum^n h_i v_i = 2 \sum^n \Delta t_i v_i^2$$

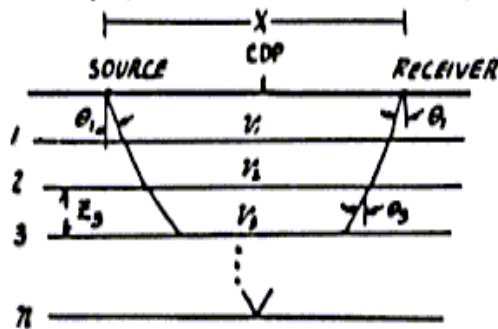
INSERTING (4) INTO (2),

$$(5) \quad t_x^2 = t_0^2 + \frac{t_0 x^2}{2 \sum \Delta t_i v_i^2}$$

$$t_x^2 = t_0^2 + x^2 / v_{rms}^2$$

This summary slide shows that the only item changing in the definition of the reflection equation is the definition of V_{nmo} .

TIME EXPRESSION - HORIZONTAL MULTILAYERS



Alternate Method: TAYLOR SERIES EXPANSION

$$T_{X,n}^2 = T_{0,n}^2 + \frac{X^2}{V_{RMS,n}^2} + \dots \text{H.O. terms.}$$

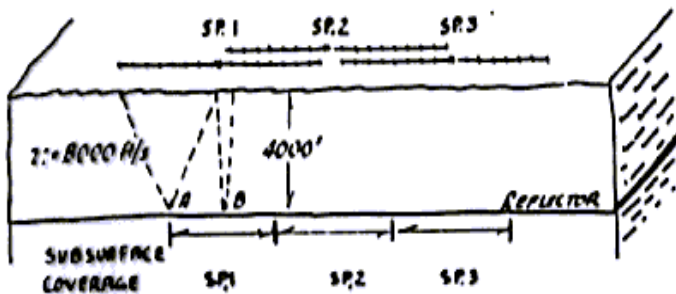
$$V_{RMS,n}^2 = \frac{\sum_{i=1}^n v_i^2 \Delta t_i}{\sum_{i=1}^n \Delta t_i}$$

SUMMARY OF VELOCITY NMO EQUATIONS

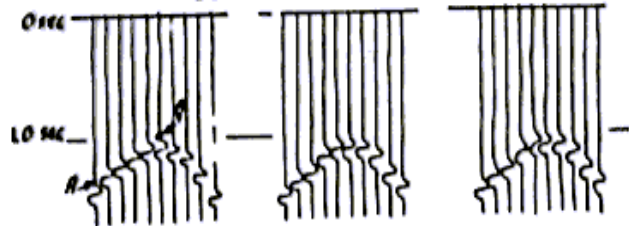
STRUCTURE		EQUATION	$V_{nmo} = f(v_i)$
	HORIZONTAL SINGLE LAYER	$T_X^2 = T_0^2 + \frac{X^2}{V_{NMO}^2}$	$V_{NMO} = \text{INTERVAL VELOCITY}$
	DIPPING SINGLE LAYER		$V_{NMO} = \frac{v_1}{\cos \alpha}$
	HORIZONTAL MULTI LAYERS		$V_{NMO} \approx V_{RMS}$ $V_{RMS}^2 = \frac{\sum_{i=1}^n v_i^2 \Delta t_i}{\sum_{i=1}^n \Delta t_i}$
	DIPPING, PARALLEL MULTI LAYERS		$V_{NMO} \approx \frac{V_{RMS}}{\cos \alpha}$
	DIPPING, NONPARALLEL MULTI LAYERS		ITERATIVE RAY TRACING PROBLEM

Schematic designed to give information on the sensitivity of normal move out to velocity, depth, and offset.

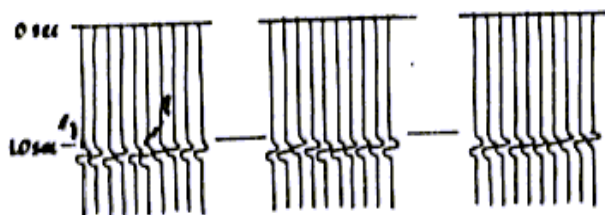
PHYSICAL SIGNIFICANCE OF NORMAL MOVEOUT



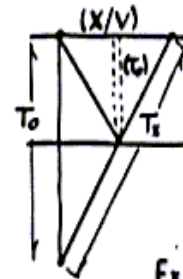
SEISMIC FIELD DATA



(+ NORMAL MOVEOUT CORRECTION)



ANALYTIC EXPRESSION OF NMO - No Dip
NMO = TIME CORRECTION DUE TO SHOT-RECEIVER SEPARATION.



$$T_x^2 = T_0^2 + (X/V)^2$$

$$T_x \triangleq T_0 + \Delta t$$

NMO CORRECTION

EXACT EXPRESSION OF NMO

$$\Delta t = \sqrt{T_0^2 + (X/V)^2} - T_0$$

APPROXIMATE EXPRESSION OF NMO

$$(T_0 + \Delta t)^2 = T_x^2 = T_0^2 + (X/V)^2$$

$$T_0^2 + 2T_0\Delta t + \Delta t^2 = T_0^2 + (X/V)^2$$

neglect

$$\Delta t = \frac{1}{2T_0} \left(\frac{X}{V} \right)^2$$

- NOTE: (1) NMO proportional to offset squared
(2) NMO inversely proportional to velocity squared.
(3) NMO inversely proportional to T_0 (depth).



Display and discussion of field acquisition geometry

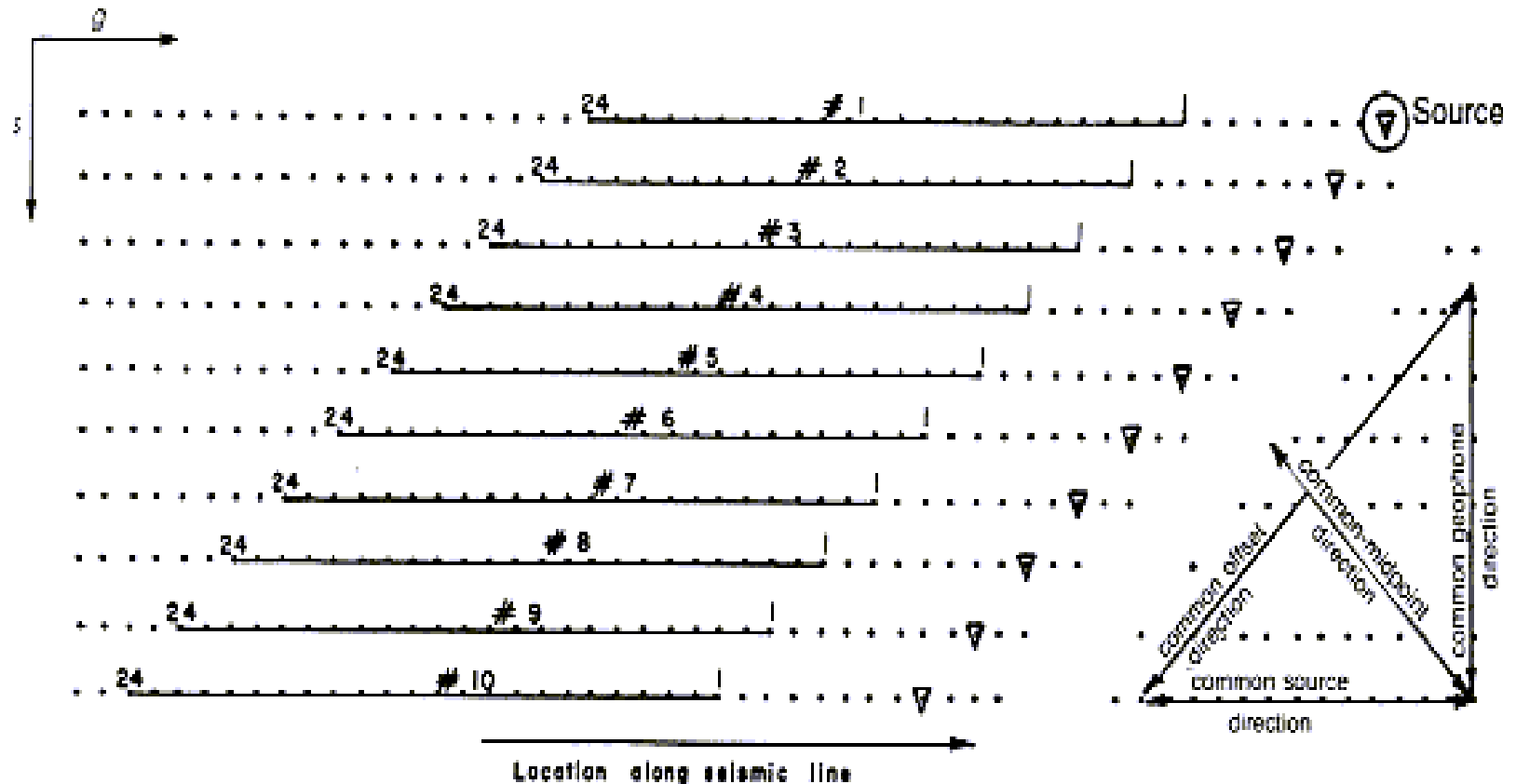
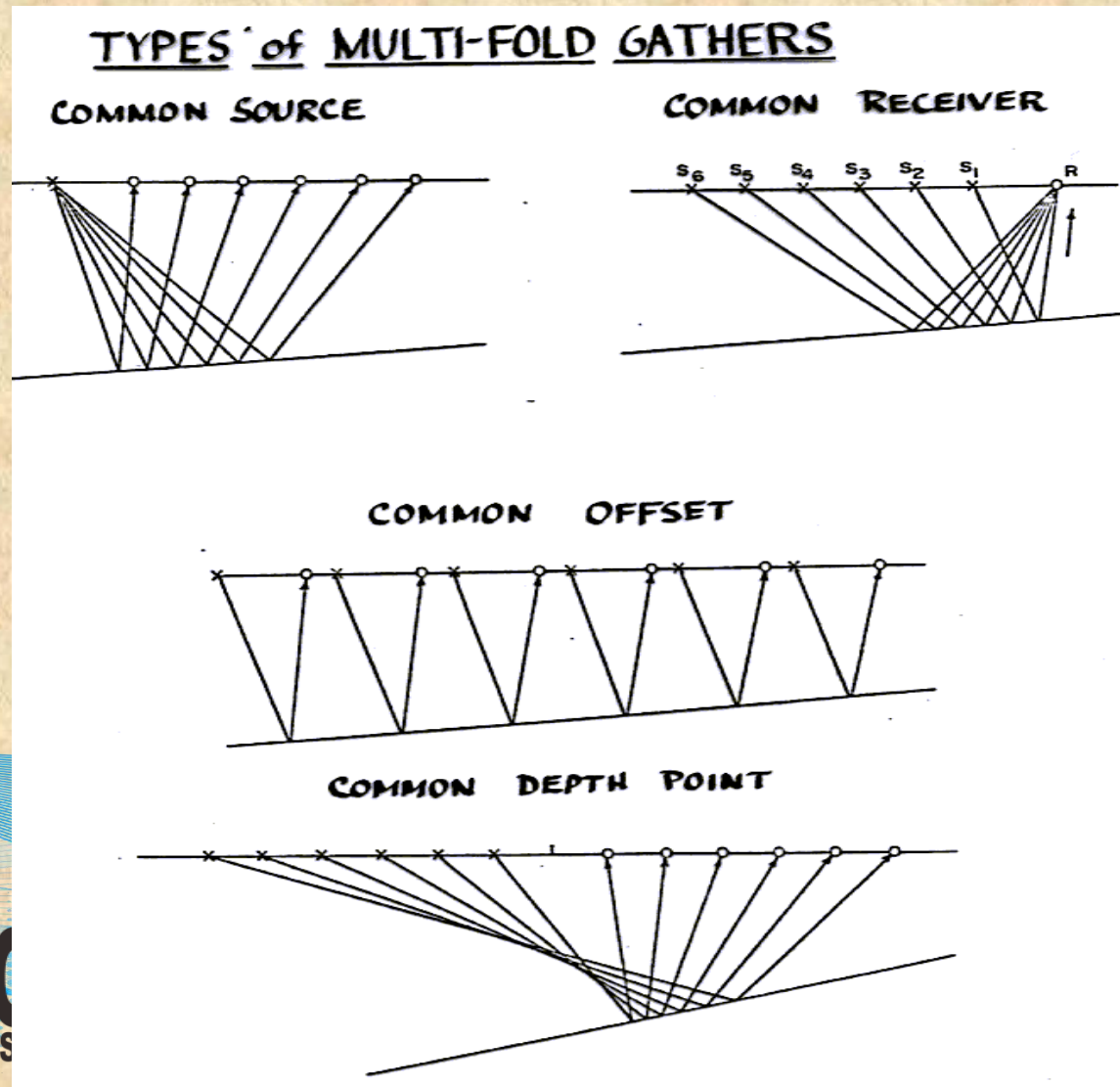
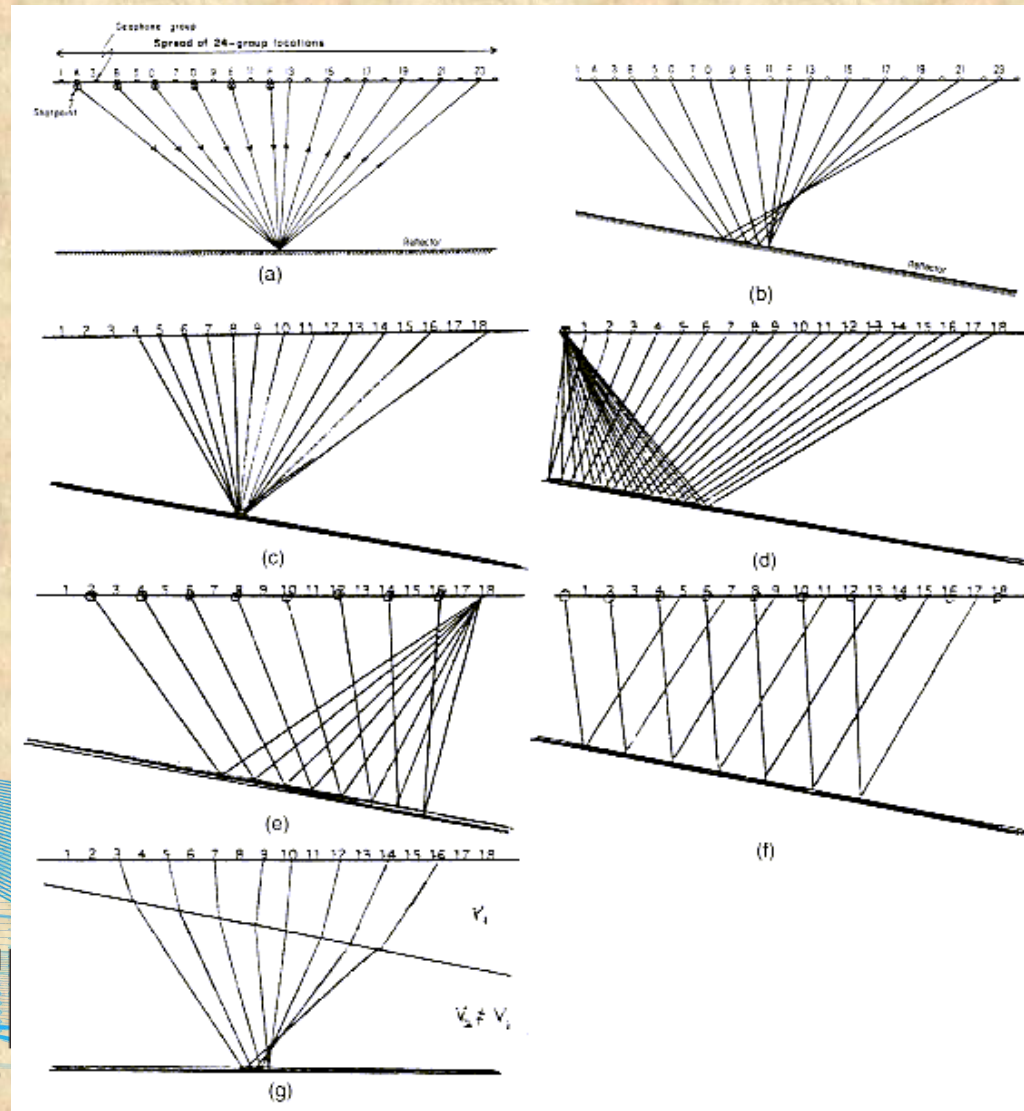


FIG. S-20. Surface stacking chart. Each horizontal line shows the location of geophone groups and source point for a single record. To the right is shown the directions for finding all traces with various elements in common, s = source coordinate, g = geophone coordinate.

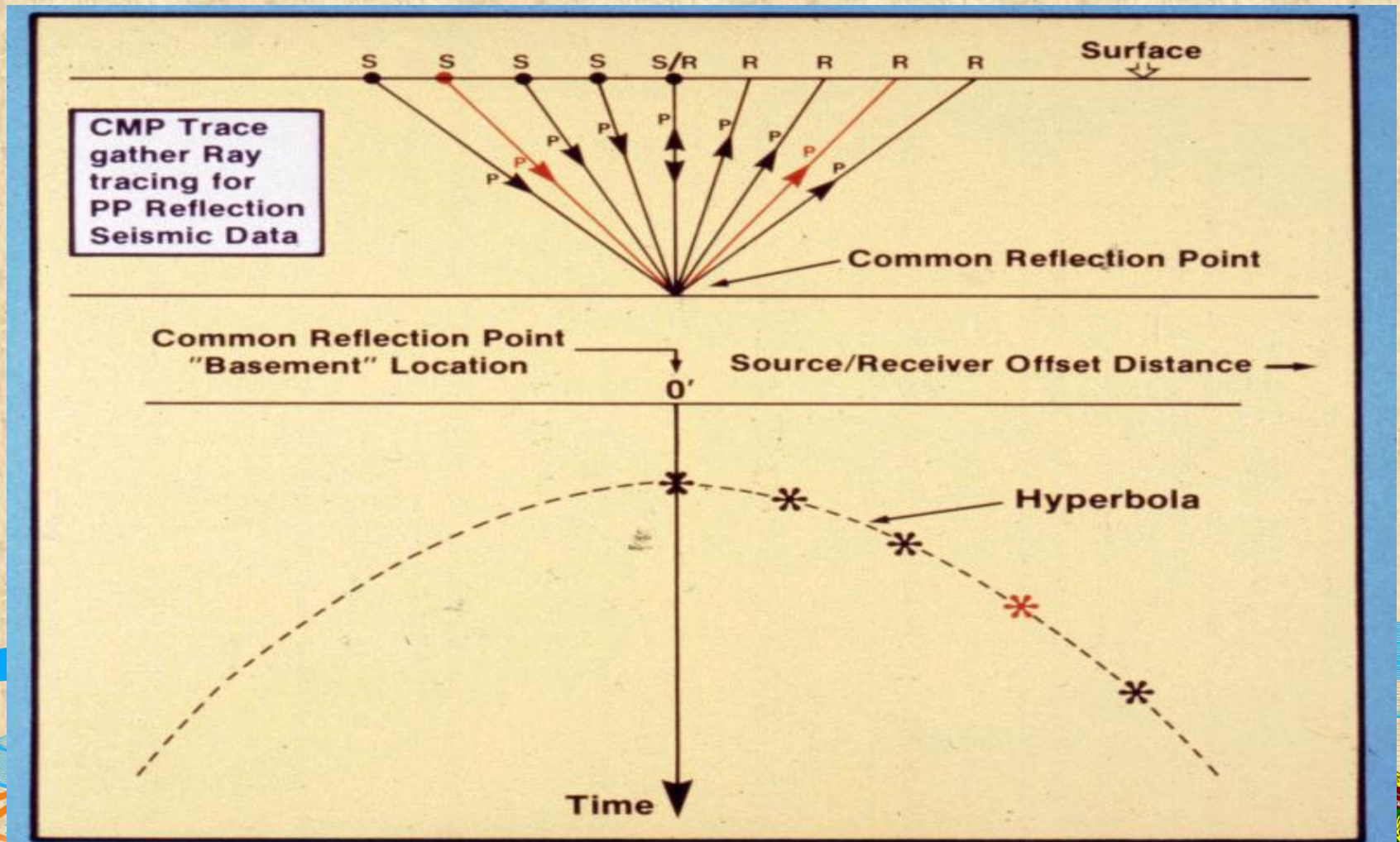
Schematic of the variety of gathers of traces that we will be discussing with a diagram to give some indication of their origin.



Schematic of the variety of gathers of traces that we will be discussing with a diagram to give some indication of their origin.



First...the trivial for this audience – the **PP** CDP gather



Snell's Law for incident P-waves and S-waves

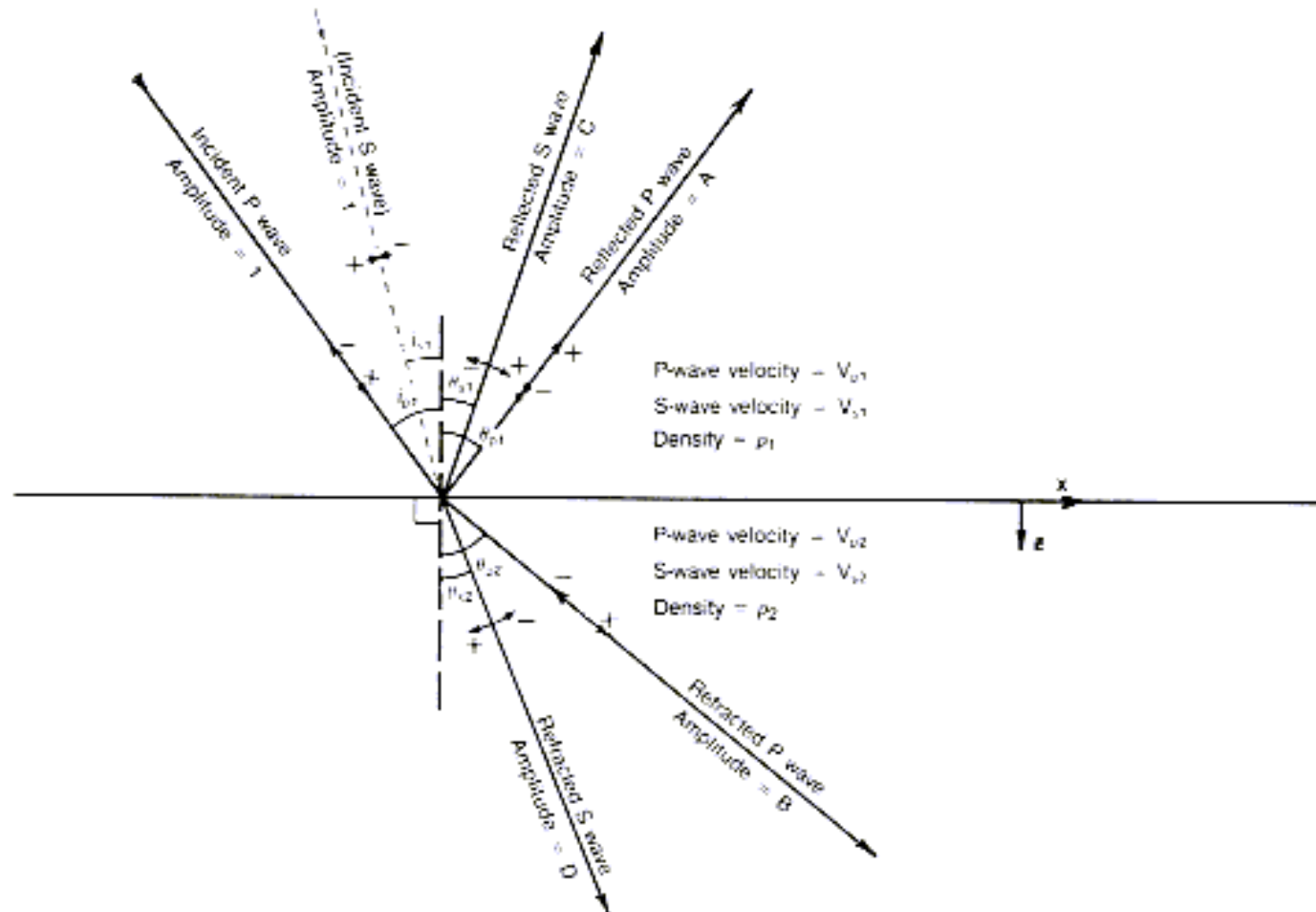
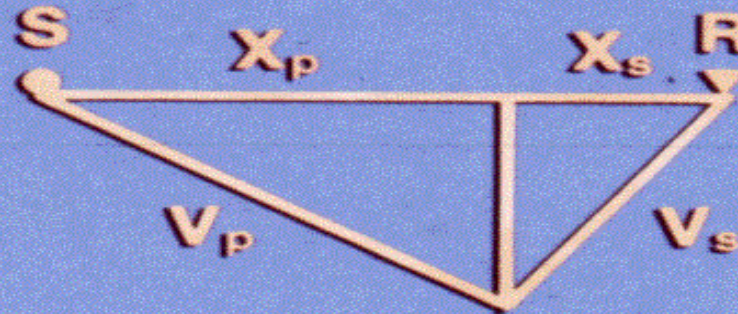


FIG. S-11. Snell's law relations for either incident P-wave or S-wave: $V_1/\sin i = V_{p1}/\sin \theta_{p1} = V_{s1}/\sin \theta_{s1} = V_{p2}/\sin \theta_{p2} = V_{s2}/\sin \theta_{s2} = 1/p$, where p is the raypath parameter.



The normal moveout equation for mode-converted shear waves has terms for the compressional and shear parts.

Normal Move Out

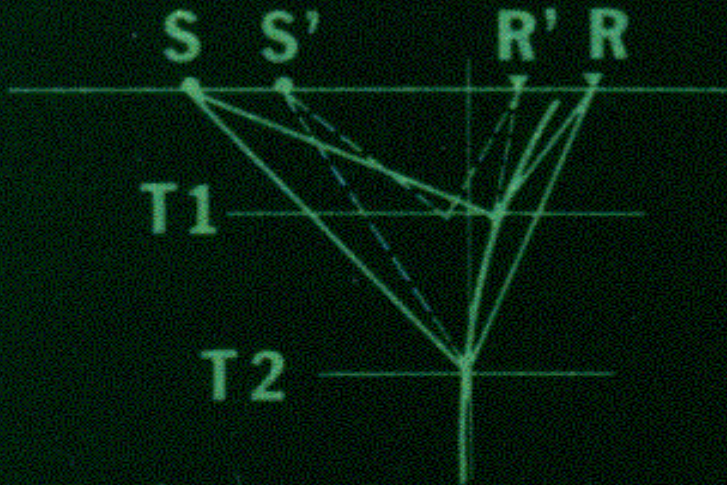


$$T = T_p + T_s$$

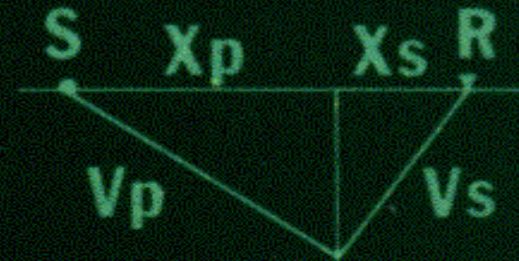
$$\Delta T = \left(-T + \sqrt{T_p^2 + \frac{X_p^2}{V_p^2}} + \sqrt{T_s^2 + \frac{X_s^2}{V_s^2}} \right)$$

For mode-converted shear waves, the normal moveout equation has terms associated with both body-wave velocities.

CDP GATHER

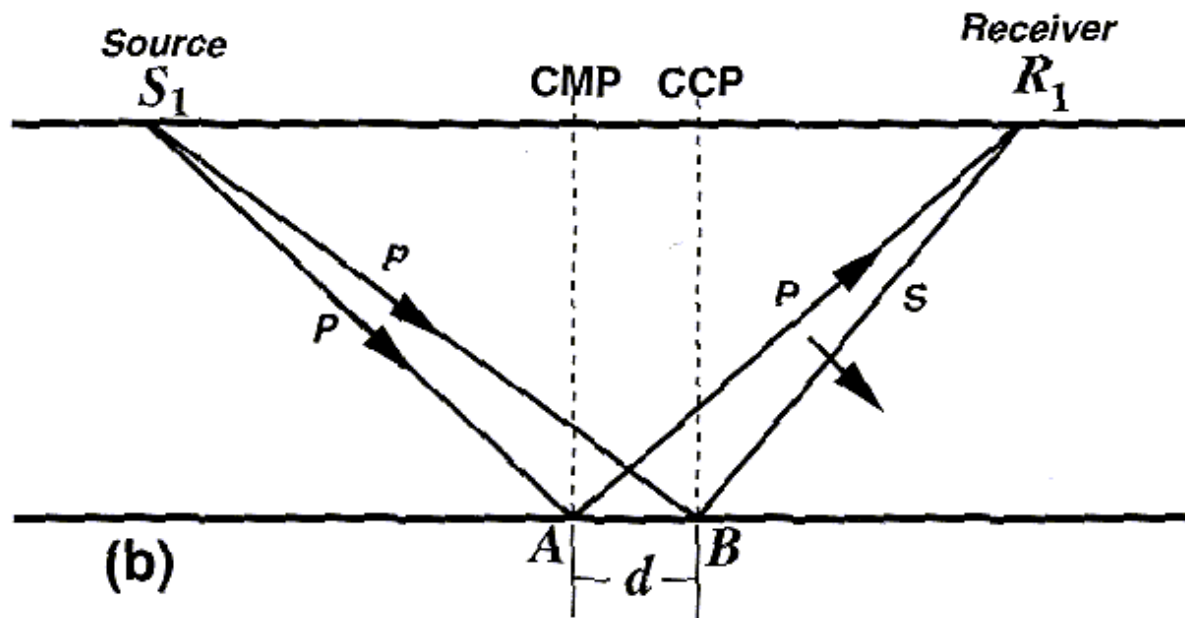
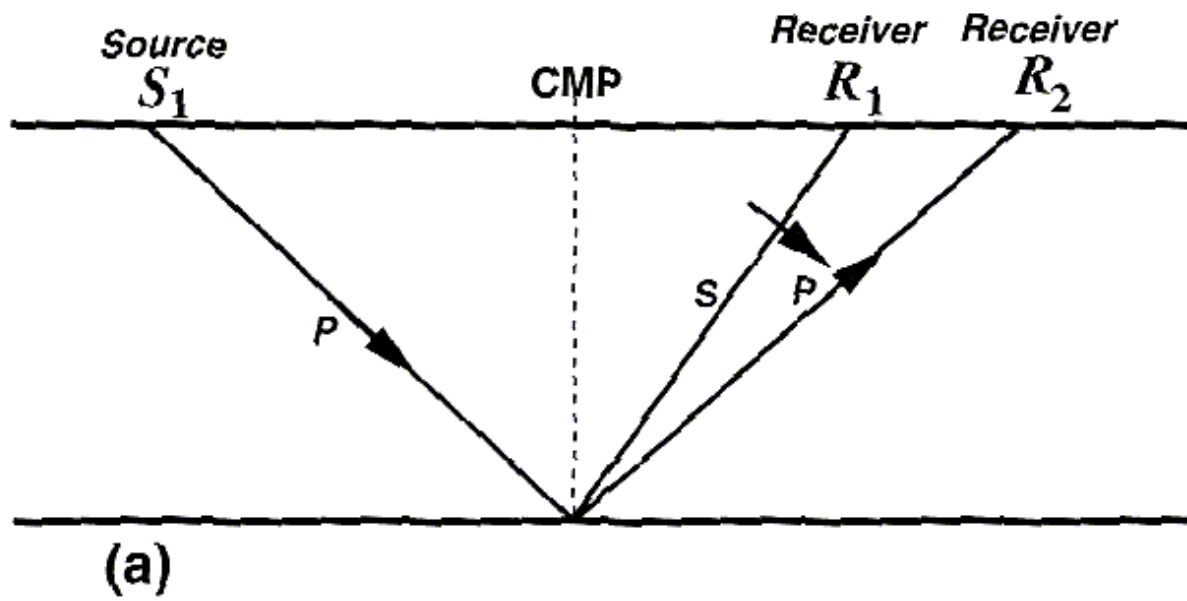


NORMAL MOVE OUT



$$T = T_p + T_s$$

$$\Delta T = \left(-T + \sqrt{T_p^2 + \frac{X_p^2}{V_p^2}} + \sqrt{T_s^2 + \frac{X_s^2}{V_s^2}} \right)$$



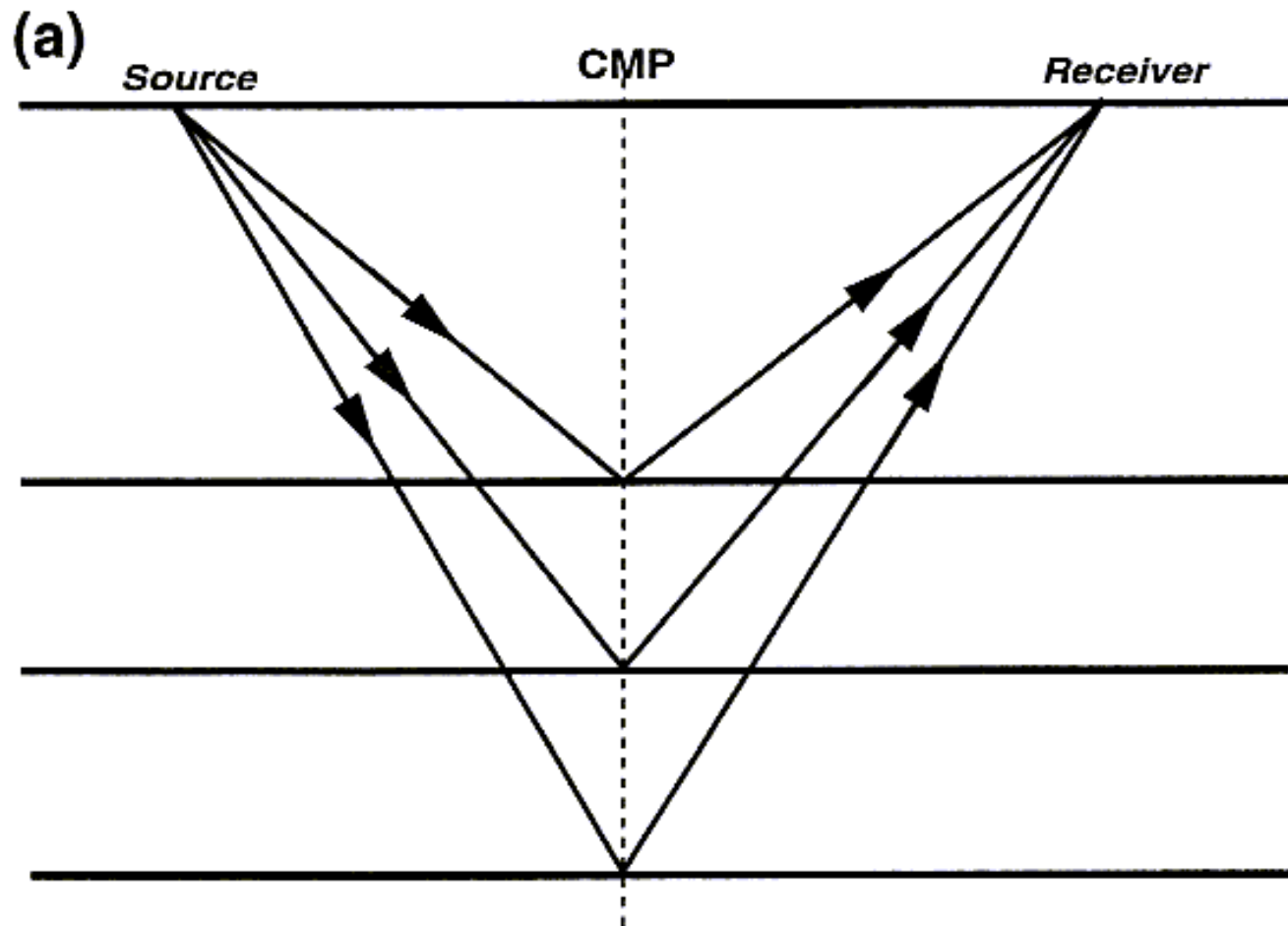


FIG. 11.6-31. Geometry of (a) common-midpoint (CMP) raypaths, and (b) common-conversion-point (CCP) and raypaths. At infinite depth, the CCP location reaches an asymptotic conversion point (ACP). See text for details.

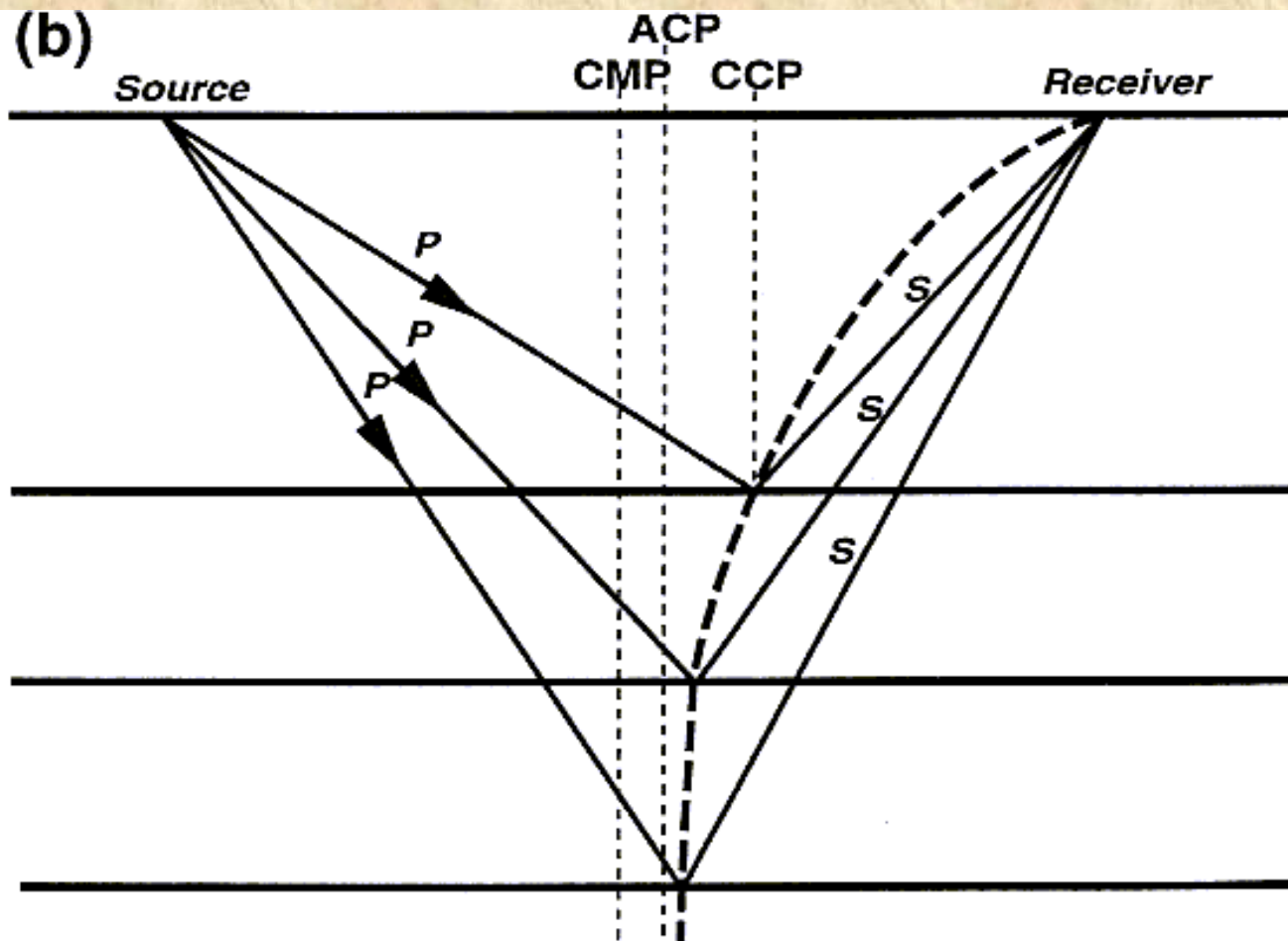


FIG. 11.6-31. Geometry of (a) common-midpoint (CMP) raypaths, and (b) common-conversion-point (CCP) and raypaths. At infinite depth, the CCP location reaches an asymptotic conversion point (ACP). See text for details.

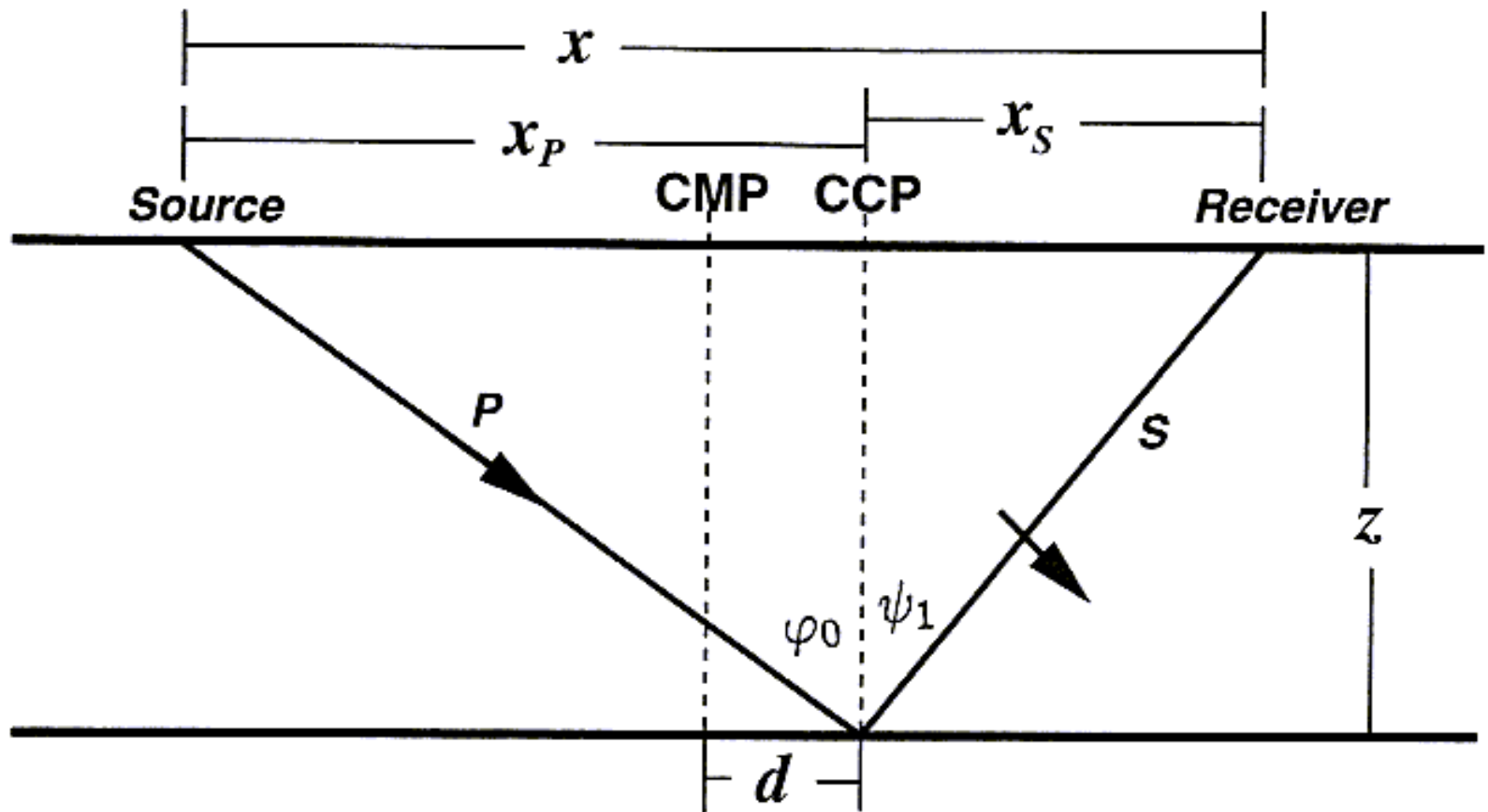
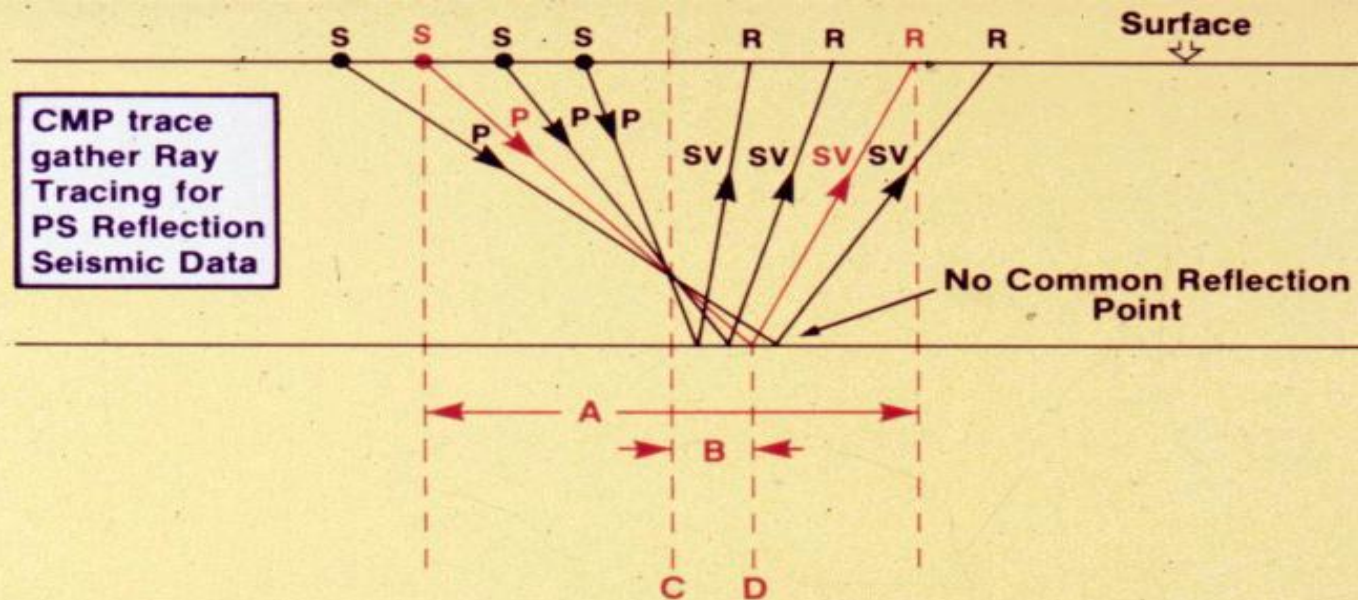


FIG. 11.6-32. Geometry of a common-conversion-point (CCP) raypath used to derive the reflection traveltime equation (11-74) for the PS -wave.

So now...the **PS** "CDP" gather **if** you sorted as **PP**



Distance A = offset between source and receiver fixed by survey

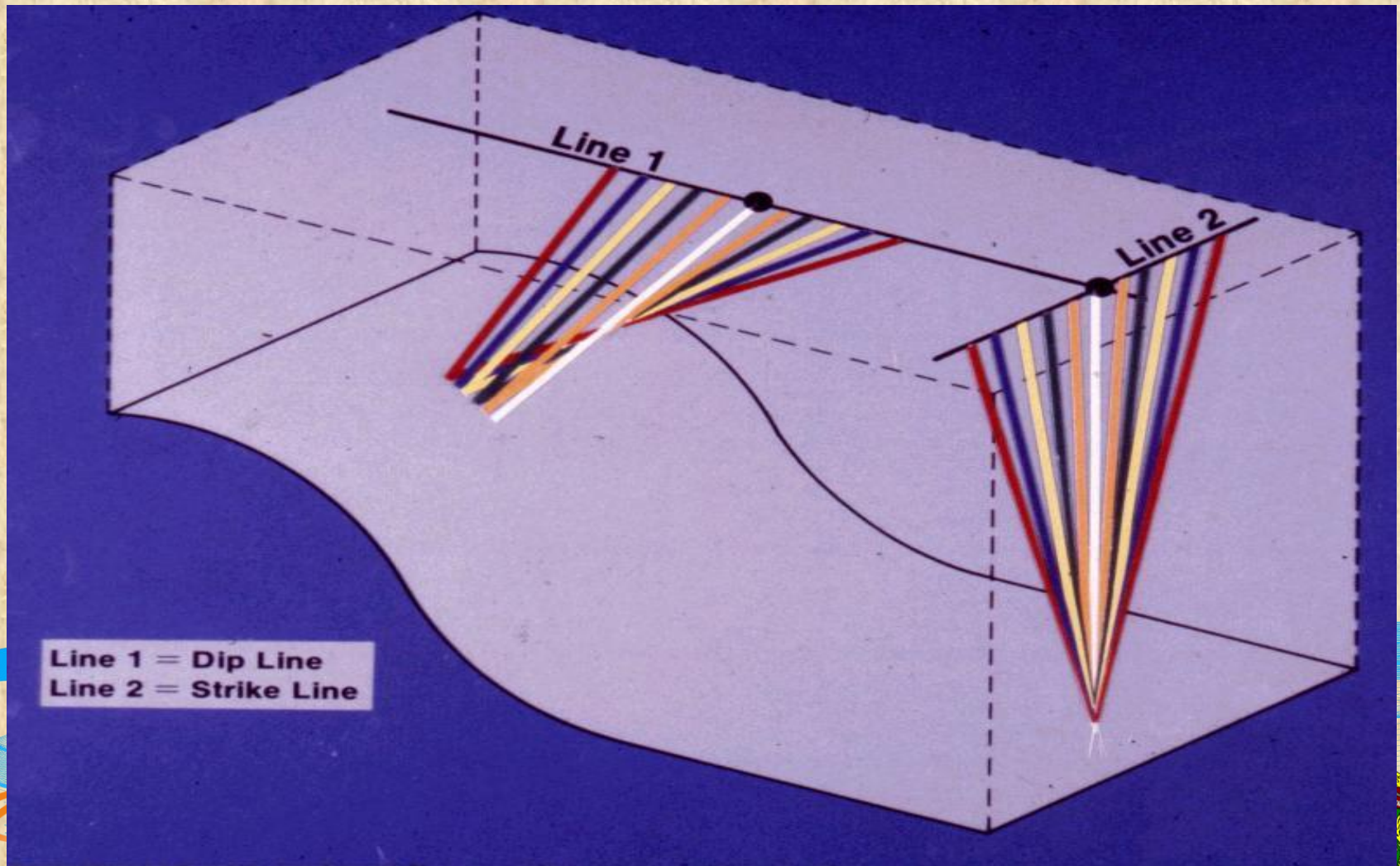
Distance B = variance from common-mid-point governed by V_P / V_S of layer

Location C = common mid point (assumed reflection point in conventional stack)

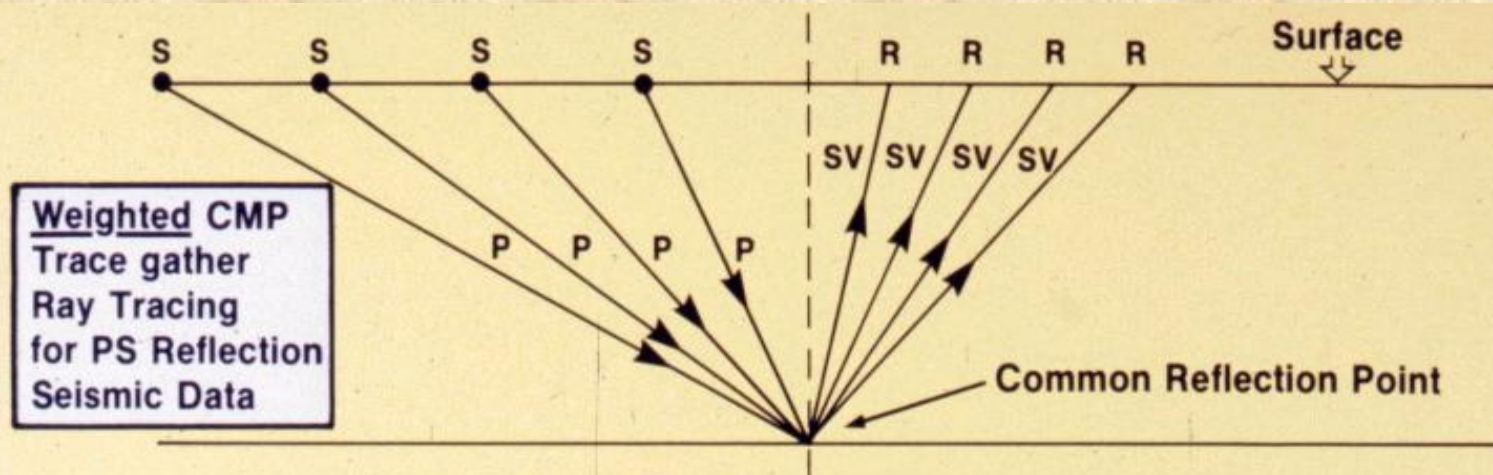
Location D = true reflection point for P / SV converted wave

Example: if $V_P / V_S = 2$, then "Basement" header word needs to be changed by $B = \frac{1}{6} A$.

What if you do not make the adjustment?...Recall **PP** CDP smear in steep-dip area...so, for a first approximation...**PS** smear "fine"



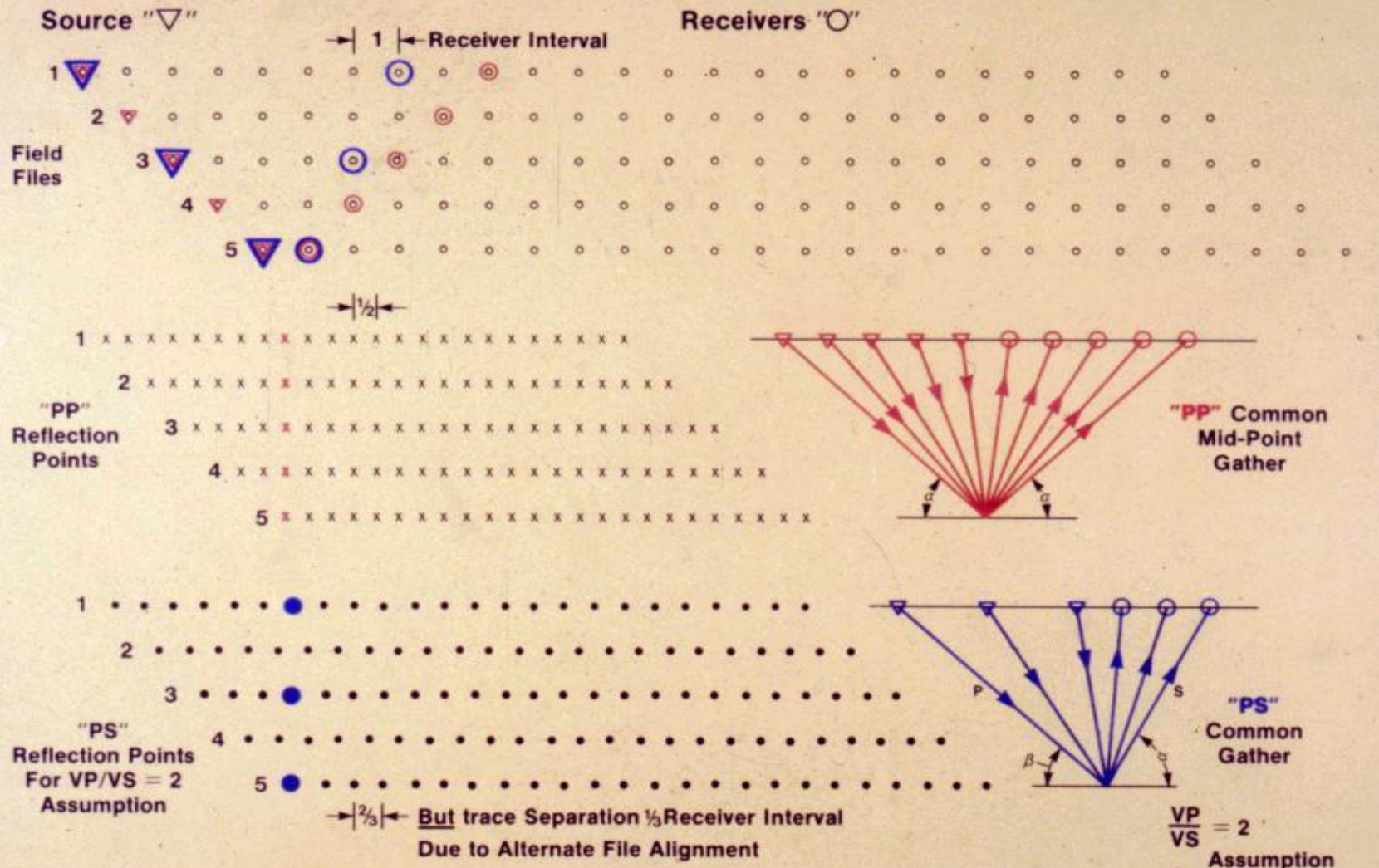
Some adjustments can be made to the **PS** processing



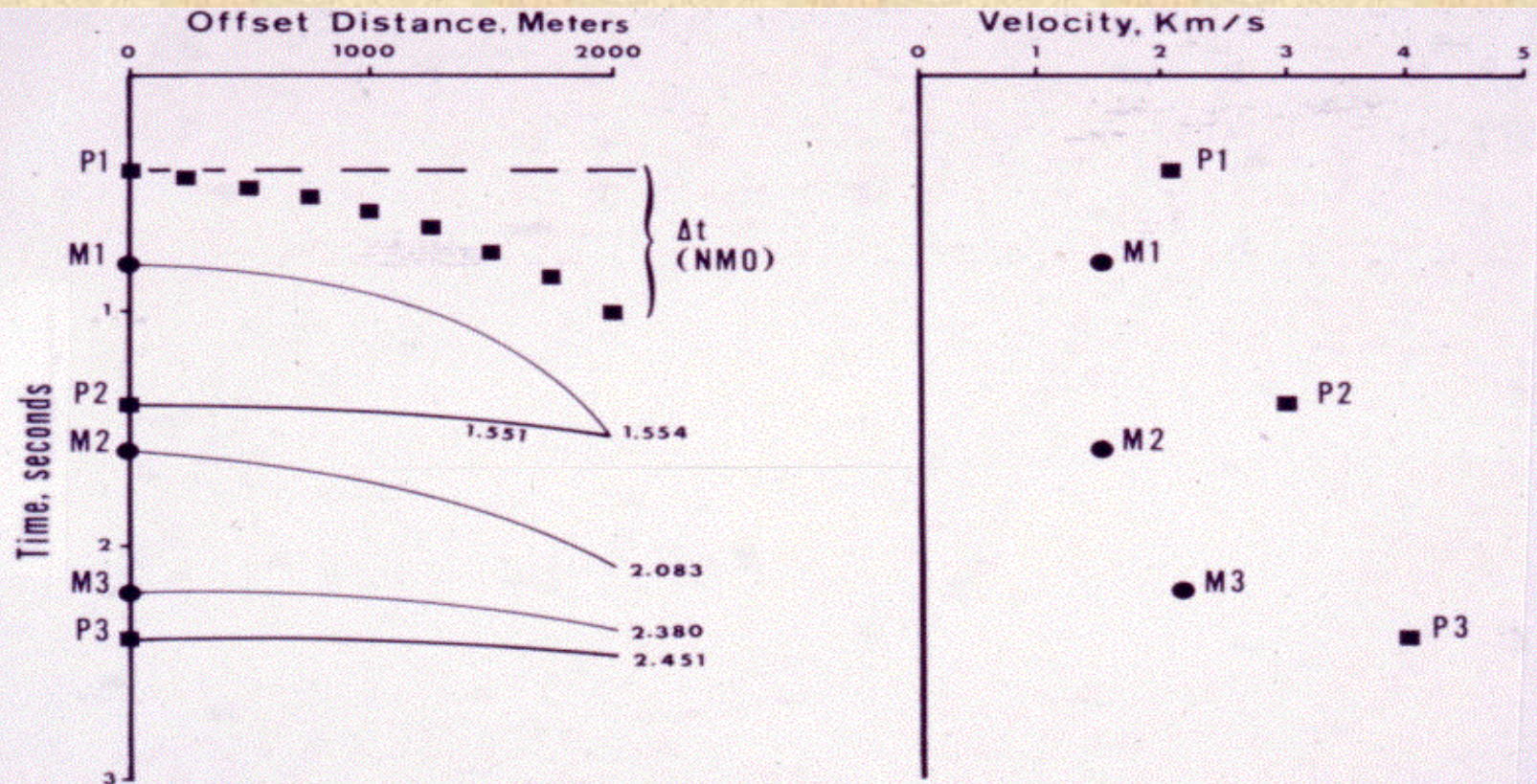
Example Comparison of CMP and Weighted CMP Data Processing

Parameter	CMP	WCMP	Discussion
Receiver Interval	165'	165'	Data Collection done similarly
# of Receivers	48	48	Data Collection done similarly
"Basement" Interval	82.5'	55'	Assumption: $VP/VS = 2$
Max Fold Stack	24	16	Assumption: $VP/VS = 2$
# Stacked Traces	100	150	Assumption: $VP/VS = 2$
Incidence/Reflection Δ	equal	unequal	Due to mode conversion on reflection
Reflection Moveout	hyperbolic	$\approx$ hyperbolic	Consequence of weighted CMP collection
Receiver Element	vertical	radial	To receive reflected transverse waves

Here are some more comparisons between PP and PS

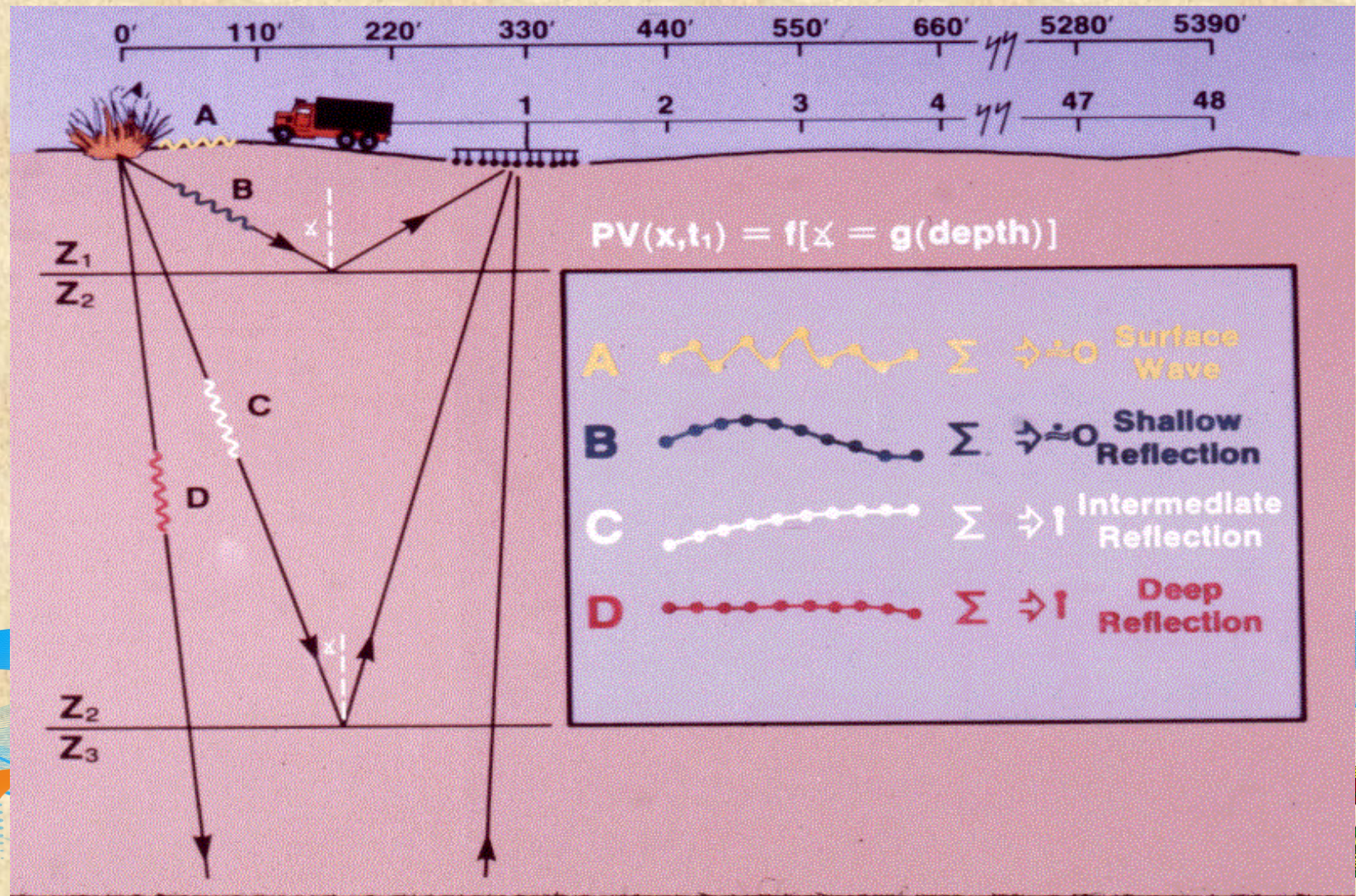


Because of different NMO and T0 for simple multiples, they can be identified and discriminated against in velocity analysis.

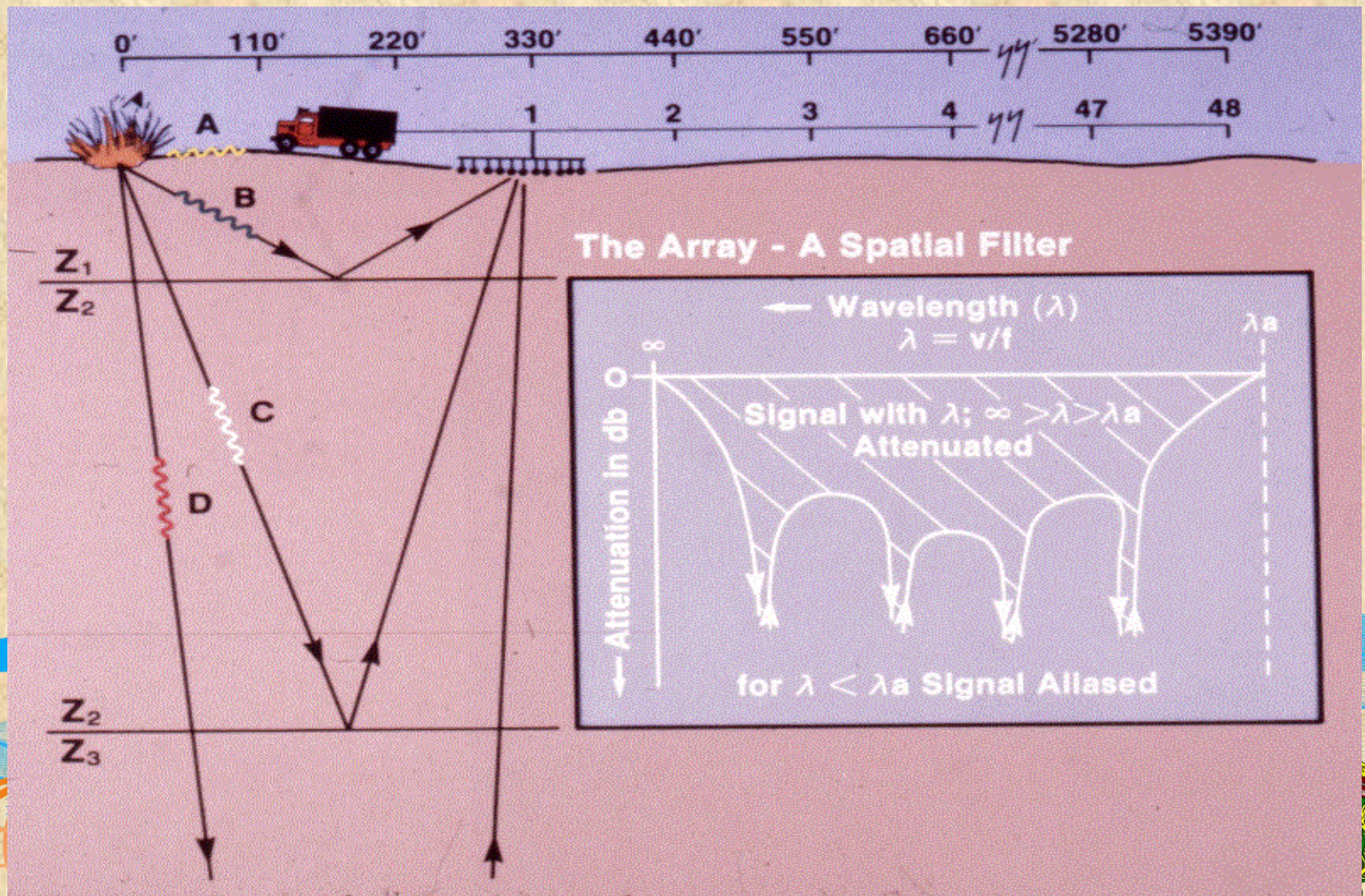


**Discrimination Between Primary
and
Multiple Seismic Reflection Events**

Arrays of geophones used to cancel horizontally-traveling 'noise' such as ground roll, while enhancing deep, vertically-traveling reflections, can be detrimental to shallow reflections.



Arrays of receiver or source points designed to eliminate horizontally traveling energy also attenuate high-frequency content from reflections.



Schematics and discussion associated with discriminating effects of geophone arrays deployed to reduce surface-wave events.

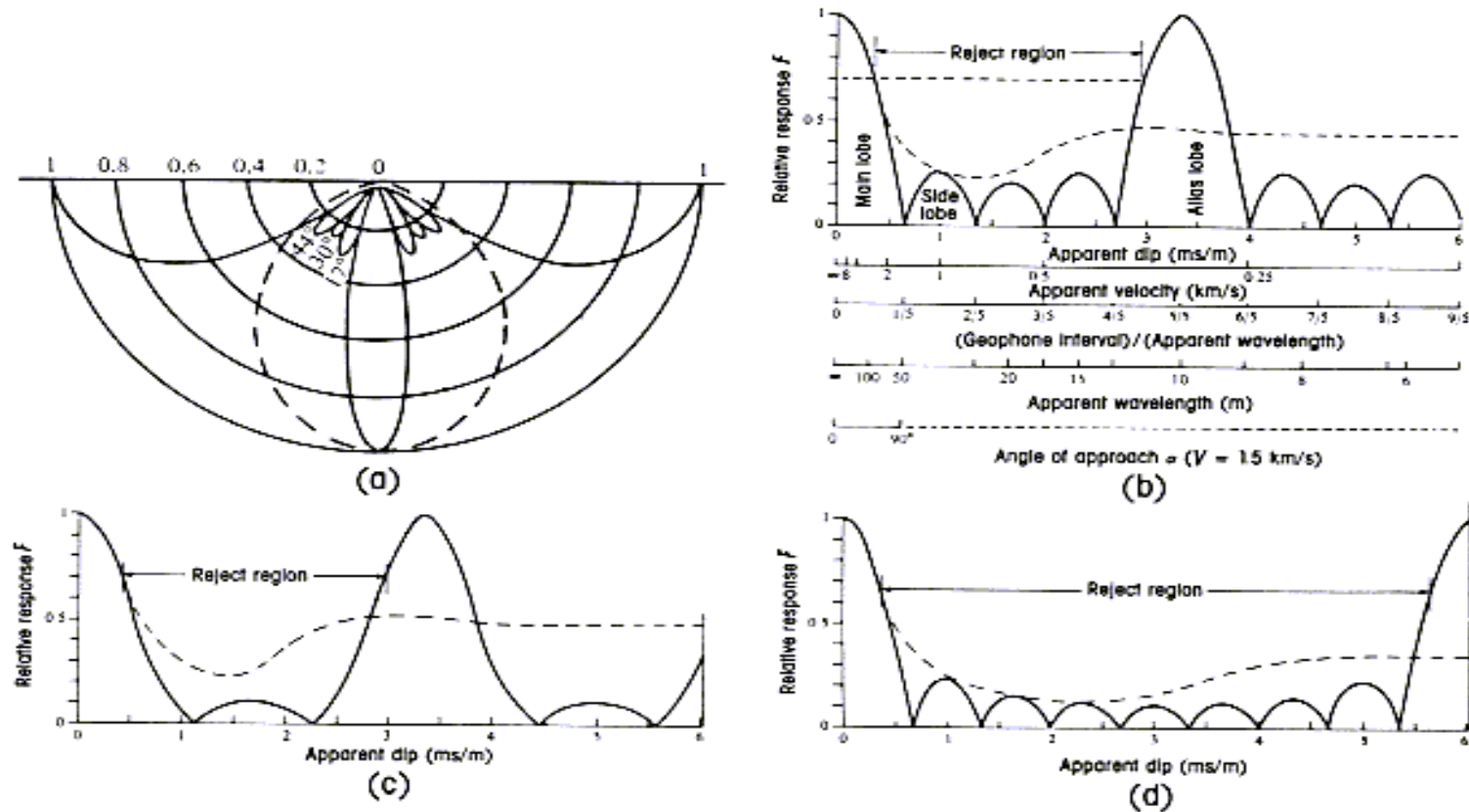
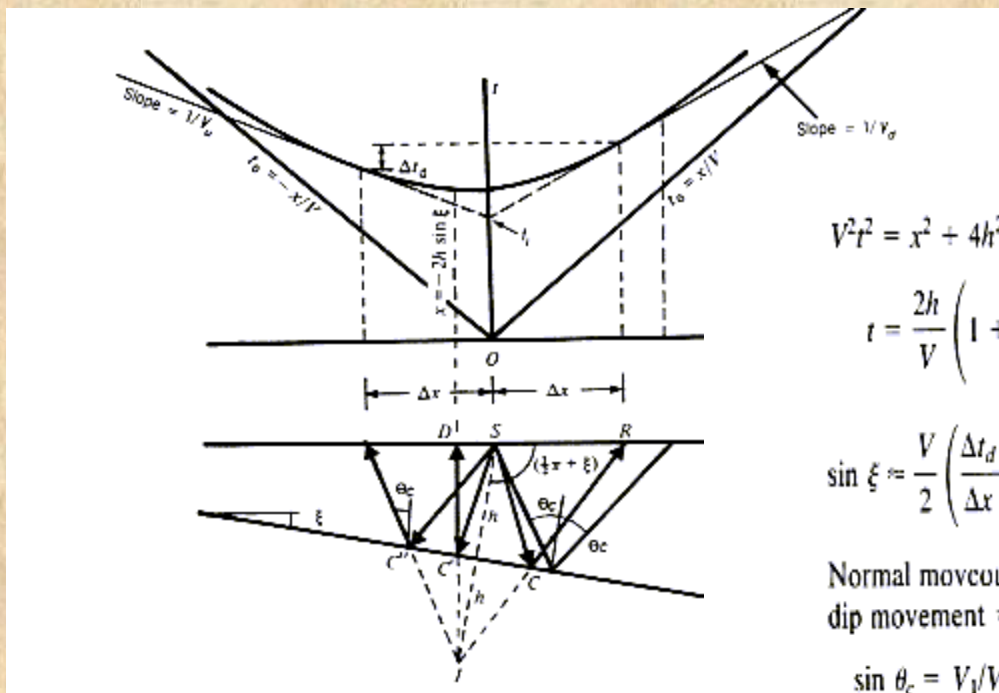


FIG. D-12. **Directivity graphs.** (a) Polar plot showing the relative amplitude of a radiated wave (or the relative sensitivity to waves approaching from different directions). (b) Directivity of five in-line geophones spaced 10 m apart. (c) Response of a tapered array of five geophones spaced 10 m apart and weighted 1,2,3,2,1; such weighting could be achieved with nine geophones distributed as the weighting. (d) Response of nine geophones spaced 5.5 m apart. The solid curves are for harmonic (steady-state) waves, the dashed curves for a transient with a bell-shaped spectrum peaked at 30 Hz and a width of 30 Hz. (From Sheriff and Geldart v. I, 1982, p.141–142.)

Reference equations and diagrams for refraction and reflection



$$V^2 t^2 = x^2 + 4h^2 + 4hx \sin \xi;$$

$$t = \frac{2h}{V} \left(1 + \frac{x^2 + 4hx \sin \xi}{4h^2} \right)^{1/2} \approx t_0 \left(1 + \frac{x^2 + 4hx \sin \xi}{8h^2} \right);$$

$$\sin \xi \approx \frac{V}{2} \left(\frac{\Delta t_d}{\Delta x} \right); h = \frac{1}{2} V t_0.$$

$$\text{Normal moveout} = \Delta t_n = (x^2/2V^2t_0) \{1 - (x/4h)^2 + \dots\}, \xi = 0;$$
$$\text{dip movement} = \Delta t_d / \Delta x.$$

$$\sin \theta_c = V_1/V_2, \quad V_1 < V_2;$$

$$V_d = V_l / \sin(\theta_c + \xi); V_u = V_l / \sin(\theta_c - \xi);$$

$$t_f = (2h/V_1) \cos \theta_c;$$

$$\theta_r = \frac{1}{3} \{ \sin^{-1}(V_l/V_d) + \sin^{-1}(V_l/V_u) \}; \quad \xi = \frac{1}{3} \{ \sin^{-1}(V_l/V_d) - \sin^{-1}(V_l/V_u) \}$$

$$V_2 = 2[(1/V_d) + (1/V_u)]^{-1} \approx \frac{1}{2}(V_d + V_u);$$

$$\xi \cos \theta_c \approx 1 - V_\phi/V_2 = V_{\phi c}/V_2 = 1.$$

FIG. R-7. Reflection and refraction equations for constant velocity overburden.